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From Reactive Vision-Language-Action Policies to Predictive World Models: A Review for Robotic Manipulation

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ABSTRACT Vision-language-action (VLA) policies are reshaping robotic manipulation by enabling robots to translate visual observations and linguistic instructions into actions. However, strong benchmark performance does not establish whether these systems can anticipate the consequences of their actions or remain dependable beyond familiar conditions. This gap is driving a shift from predominantly reactive policies toward predictive systems that model how the environment may evolve under different actions. This article critically examines that shift through a synthesis of 147 reports on VLA policies, world models, and evaluation resources. We identify seven interrelated limitations across policy formulation, learning, and evaluation. We then assess the extent to which predictive mechanisms can address these limitations. The evidence shows substantial promise but remains uneven. Auxiliary predictive training and synthetic trajectory generation provide the clearest demonstrated benefits. In contrast, evidence for planning or reinforcement learning within learned world models remains narrow and task-specific. Core capabilities such as explicit comparison of candidate actions, physically valid prediction, and consequence-aware safety have yet to be demonstrated convincingly. World models should therefore be viewed not as a universal solution but as a predictive foundation whose value depends on effective action conditioning and rigorous evaluation. This review serves the robot learning and embodied artificial intelligence communities by connecting advances in VLA policies with research on predictive control and dependable autonomy. We conclude with a research agenda for verified imagination, in which learned dynamics are integrated with runtime monitoring, formal constraints, and task-and-motion planning to support reliable robotic manipulation.

INDEX TERMS Embodied artificial intelligence, model-based reinforcement learning, robot learning, robotic manipulation, systematic mapping review, video prediction, vision–language–action models, world models.

I. INTRODUCTION

ROBOTIC manipulation has advanced rapidly through vision-language-action (VLA) policies that translate visual observations and language instructions into robot actions. RT-2 established this paradigm by adapting a web-scale vision-language model to robotic data [1]. OpenVLA later provided an open implementation trained on the Open X-Embodiment corpus [2], [3]. Recent systems such as π_0, GR00T N1, Gemini Robotics, and Figure Helix have extended the approach to larger datasets and more diverse embodiments [4]–[7]. Several VLA systems now report success rates above 97% on standard LIBERO configurations [8], [9]. These results demonstrate substantial progress under the evaluated protocols. However, they do not establish whether the same policies can remain reliable when the environment or task differs from the training distribution.

World models offer a complementary approach. A world model learns how an environment changes over time and how those changes depend on an agent's actions. It can support future-state prediction, imagined training, or plan-

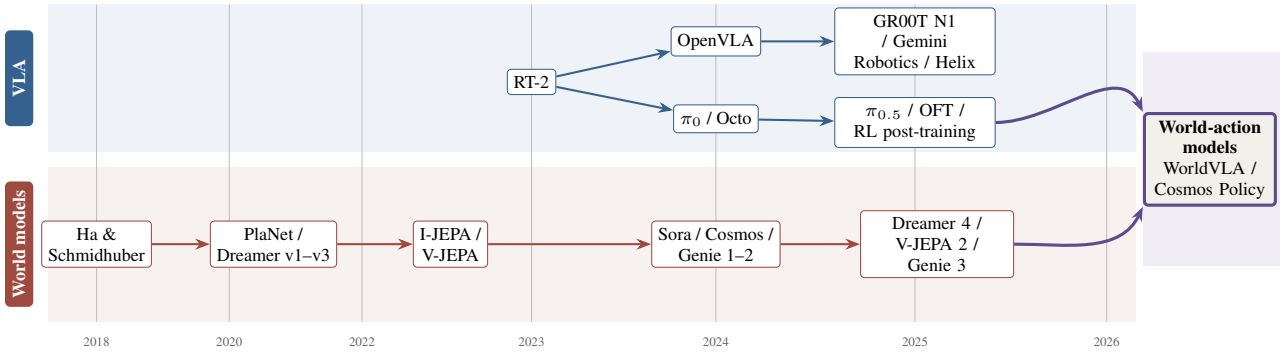


FIGURE 1. Dual-lane timeline of the field, 2018–2026 (year axis nonlinear for legibility). The VLA lane (top) industrializes reactive policies from RT-2 through $\pi_{0.5}$, optimized decoding, and reinforcement-learning (RL) post-training; the world-model lane (bottom) develops predictive models from latent dynamics through world foundation models. The lanes converge in 2025–2026 on unified world-action models (Sec. VI-G).

ning before physical execution. These capabilities distinguish explicit predictive reasoning from a predominantly reactive policy interface. In this review, reactive describes a policy that selects actions without exposing an action-conditioned transition model for comparing their predicted consequences. This definition does not imply that the policy lacks memory or an internal representation of dynamics. Reactive control and explicit planning instead represent different points along a continuum of predictive capability. Fig. 1 traces the development of VLA policies and world models, together with their recent convergence into world-action systems.

The motivation for this convergence extends beyond benchmark performance. Perturbation studies show that VLA policies can degrade substantially under changes in layout, viewpoint, or instruction phrasing [9]. Real-to-simulation studies also question whether success on common benchmarks reliably predicts physical performance [10]. Reinforcement-learning post-training has produced large improvements over supervised policies in specific LIBERO settings [11], [12]. However, these experiments change the training data and optimization process together, which makes the source of improvement difficult to isolate. Large teleoperation datasets further demonstrate the value of action-grounded experience while highlighting its acquisition cost [4]. Together, these findings expose a broader limitation: selecting actions from learned correlations is not the same as predicting their physical effects.

Predictive models could help address this limitation by allowing a robot to evaluate how alternative actions may change the environment. They could also provide imagined experience for policy learning. Yet prediction alone is not sufficient. A visually plausible future may violate contact dynamics or omit information that matters for control. Predictions may also become unreliable when actions fall outside the model's training support. A world model should therefore be treated as a candidate mechanism rather than a universal solution. Its value must be established through improvements in control and physical reliability, not through prediction quality alone.

Recent systems illustrate both the promise and the uncertainty of this direction. V-JEPA 2 combines large-scale video pretraining with action-conditioned adaptation for robotic manipulation [13]. DreamGen uses generated trajectories to expand the behavioral coverage of a robot policy [14]. Dreamer 4 demonstrates complex policy learning inside a learned environment, although its principal evidence comes from Minecraft rather than robotic manipulation [15]. WorldVLA jointly generates future video and robot actions [16], while Cosmos Policy adapts a video world model for direct control and rollout-based planning [17]. These developments indicate a movement toward unified world-action models. However, current evidence remains narrow and often combines several system components. It does not yet establish a general advantage for predictive planning in manipulation.

This review examines the transition from VLA policies to predictive systems through a systematic mapping of 147 reports. Rather than asking which architecture performs best, we focus on the relationship between limitations and proposed solutions. We first identify the major limitations affecting current VLA systems. We then examine which world-model mechanisms could address them. Finally, we assess whether the available evidence demonstrates the intended benefit. This structure separates evidence that a limitation exists from evidence that a mechanism changes it. It also distinguishes both from downstream task success.

Existing surveys provide broad accounts of VLA architectures and their deployment challenges [18]–[21]. A parallel literature reviews world models for robot learning and embodied artificial intelligence [22]–[24]. Recent studies also survey the emerging world-action-model paradigm [25], [26]. The present article does not seek to reproduce these architectural taxonomies. Instead, it connects diagnosed VLA limitations to candidate predictive mechanisms and evaluates the strength of evidence supporting each connection. The scope is centered on robotic manipulation.

The article makes four principal contributions:

1) It develops an evidence-linked diagnosis of seven limitations in current VLA research. The analysis distinguishes limitations of policy formulation from those arising in training or evaluation.

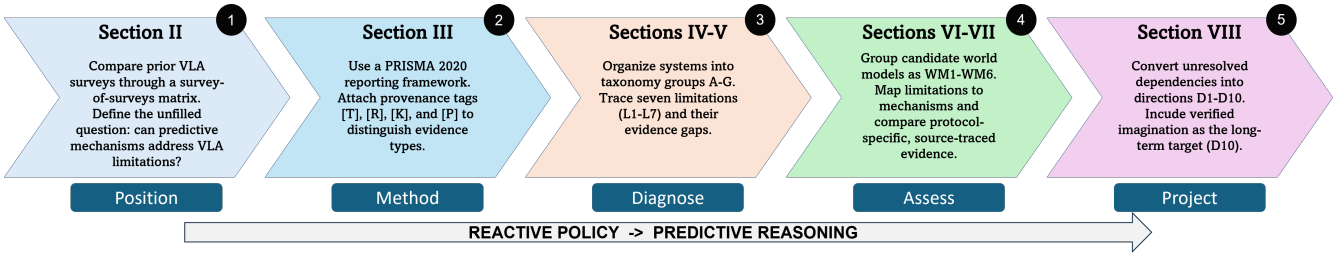


FIGURE 2. CC-CBF under Rules 13 (top) and 14 (bottom). Inside d_{act}, the barrier inflates on the COLREG-prohibited side; outside it, the isotropic radius R_b is recovered.

2) It establishes a mechanism-centered connection between VLA policies and world models. The proposed limitation-to-candidate-mechanism framework separates direct evidence from indirect or prospective claims.
3) It provides a protocol-aware synthesis of empirical findings across both research areas. Benchmark results are interpreted within their original experimental settings rather than combined into a common leaderboard. The synthesis also supports a practical framework for selecting model families under different deployment requirements.
4) It presents a testable research agenda for dependable predictive manipulation. The agenda culminates in verified imagination, which integrates learned dynamics with runtime monitoring and formal constraints. It also connects prediction with task-and-motion planning so that imagined outcomes can guide action without being treated as verified facts.

The remainder of the article follows this argument. Section II positions the review relative to existing surveys, and Section III presents the mapping methodology. Sections IV and V characterize VLA systems and diagnose their limitations. Sections VI and VII examine predictive mechanisms and evaluate the available evidence. Section VIII develops the research agenda, and Section IX concludes the article. Fig. 2 summarizes this progression.

II. RELATED SURVEYS AND POSITIONING

Surveys of VLA policies and world models have largely developed as separate literatures. VLA reviews typically focus on policy design and deployment, while world-model reviews emphasize predictive architectures and their roles in learning or planning. Table 1 compares the studies most closely related to this review.

A. SURVEYS OF VISION-LANGUAGE-ACTION MODELS

The rapid growth of VLA research has produced surveys that examine the field from architectural, representational, and deployment perspectives. Ma et al. organize VLA systems according to their components and control hierarchies [18]. Zhong et al. focus on action tokenization across discrete and continuous representations [19]. Kawaharazuka et al. examine real-world deployment, with particular attention to hardware integration and data collection [20]. Other reviews address general VLA concepts [27], large-VLM manipulation policies [28], efficient deployment [29], and reinforcement-learning post-training [30].

Din et al. present a systematic review of multimodal fusion in VLA models for robotic manipulation [21]. Their study analyzes 102 models together with major datasets and simulation platforms. It emphasizes fusion strategies, model architectures, and comparative performance. The present review addresses a different question. It examines whether predictive mechanisms can address identified limitations of VLA systems. This requires a joint analysis of VLA policies and world models rather than a taxonomy of VLA designs alone.

B. SURVEYS OF WORLD MODELS

World-model surveys have developed in parallel, with growing attention to how learned predictive representations support robot learning and planning. Hou et al. organize world models for robot learning by their function, including policy learning and learned simulation [22]. Broader reviews examine world models for embodied artificial intelligence [23], their development from language models [31], and their role in generative robotic systems [32], [33]. Li et al. provide an early account of world models for manipulation [34].

Wang et al. focus specifically on action-conditioned world models for robotic manipulation [24]. They distinguish representational and deployment roles and review evaluation methods for predictive fidelity and task performance. Their study provides a manipulation-centered world-model taxonomy. Our review complements this work by starting from limitations observed in VLA systems and assessing whether particular predictive mechanisms change the targeted outcomes.

Recent surveys by Wang et al. and Shen et al. examine the emerging world-action-model literature [25], [26]. These studies organize systems according to architecture and action coupling. They establish that the integration of prediction and control is already an active research direction. The present review does not claim to be the first account of this convergence. Its focus is the strength of evidence connecting predictive mechanisms to specific VLA limitations.

TABLE 1. Comparison with existing surveys. ✓ = covered centrally; (✓) = partial or outlook-level; — = not covered. *Lim. diag.* = structured diagnosis of VLA limitations and evidence gaps; *Lim.→mech.* = explicit limitation-to-candidate-mechanism mapping; *X-fam. synth.* = cross-family protocol-aware evidence mapping of VLAs and world models. Comparators are directly related surveys retrieved or found during the review.

Survey	Venue/Year	Primary scope	VLA	WM	Lim.	L–M	X-fam.
Ma *et al.* [18]	IEEE TNNLS 2026	VLA components/policies/planners	✓	—	(✓)	—	—
Zhong *et al.* [19]	arXiv 2025	Action tokenization	✓	(✓)	—	—	—
Kawaharazuka *et al.* [20]	IEEE Access 2025	VLA real-world deployment	✓	(✓)	(✓)	(✓)	—
Sapkota *et al.* [27]	arXiv 2025	VLA concepts & applications	✓	—	(✓)	—	—
Shao *et al.* [28]	arXiv 2025	Large-VLM manipulation policies	✓	—	—	—	—
Yu *et al.* [29]	arXiv 2025	Efficient/on-device VLAs	✓	—	(✓)	—	—
Deng *et al.* [30]	TechRxiv 2025	RL post-training of VLAs	✓	—	(✓)	—	—
Din *et al.* [21]	Information Fusion 2026	Multimodal fusion in VLA manipulation	✓	-	(✓)	-	-
Hou *et al.* [22]	arXiv 2026	World models for robot learning	(✓)	✓	—	—	—
Li *et al.* [23]	arXiv 2025	World models for embodied AI	—	✓	—	—	—
Feng *et al.* [31]	IEEE CAS Mag. 2025	Large language models (LLMs) to world models	(✓)	✓	—	—	—
Zhang *et al.* [32]	arXiv 2025	Generative AI in manipulation	(✓)	(✓)	—	—	—
Zhang *et al.* [36]	arXiv 2025	Pure VLA architectures	✓	—	(✓)	—	—
Ding *et al.* [33]	ACM CSUR 2025	World models (general)	—	✓	—	—	—
Li *et al.* [34]	arXiv 2025	World models for manipulation	(✓)	✓	—	(✓)	—
Wang *et al.* [24]	arXiv 2026	World models for manipulation	(✓)	✓	(✓)	(✓)	(✓)
Wang *et al.* [25]	arXiv 2026	World-action models	✓	✓	(✓)	—	—
Shen *et al.* [26]	arXiv 2026	World-action models (anatomy)	✓	✓	(✓)	—	—
This mapping review	—	**VLA→WM evidence map**	✓	✓	✓	✓	(✓)

C. POSITIONING OF THIS REVIEW

This review is distinguished by the combination of three elements. First, it develops an evidence-linked diagnosis of limitations affecting current VLA systems and their evaluation. Second, it maps each limitation to candidate world-model mechanisms and distinguishes demonstrated effects from prospective benefits. Third, it evaluates results across both model families under a protocol-aware evidence policy.

This distinction is important because evidence that a limitation exists does not show that a proposed mechanism resolves it. Likewise, improved task performance does not establish that prediction caused the improvement. Our analysis therefore separates limitation evidence from mechanism-effect evidence and downstream task benefit. Source traceability is also recorded independently of study validity and replication status.

The scope is centered on robotic manipulation. Work on autonomous driving and pure navigation is included only when it introduces methods that directly inform manipulation. Readers seeking broader coverage of world models for driving are referred to the specialized survey literature [35].

III. LITERATURE SELECTION AND SEARCH CRITERIA

This section summarizes the search and selection process used to identify the literature reviewed in this article. A PRISMA-style procedure [37] was followed to identify relevant studies and apply consistent screening criteria (Fig. 3). The following subsections describe the research questions, search terms, eligibility criteria, study selection, and evidence-extraction process.

A. RESEARCH QUESTIONS

The review addresses four research questions:

RQ1: Which VLA systems have been proposed for robotic manipulation, and what problems do they address?

RQ2: What limitations and evaluation gaps are documented in current VLA research?

RQ3: Which world-model mechanisms are relevant to these limitations, and how directly are their effects supported?

RQ4: Under which evaluation protocols can results be compared, and where does the evidence remain incomplete?

B. SEARCH STRATEGY

The search covered arXiv, IEEE Xplore, the ACM Digital Library, OpenReview, and proceedings from CoRL, ICRA, IROS, RSS, NeurIPS, ICLR, and CVPR. Curated research lists and benchmark leaderboards were used as additional discovery sources. The search window extended from January 2023 to August 2026 for VLA reports and from January 2018 to August 2026 for world models and evaluation resources. Earlier model-based control research was used for historical context but was not systematically screened.

Two primary query families were used, with syntax adapted to each database:

> **VLA query:** (‘vision-language-action” OR ‘VLA model” OR ‘robot foundation model” OR ‘generalist robot policy”) AND (manipulation OR grasping OR “mobile manipulation”)
>
> **World-model query:** (‘world model” OR ‘video prediction” OR ‘latent dynamics” OR ‘JEPA” OR ‘world foundation model” OR ‘world-action model” OR ‘world action model” OR ‘world-action foundation model” OR “WAM”) AND (robot* OR manipulation OR embodied)

Targeted searches were then conducted for topics not fully captured by the primary queries. These included action tokenization, flow-matching policies, predictive control, learned simulators, reinforcement-learning post-training, synthetic robot trajectories, physical-interaction prediction, and world-model evaluation. Forward and backward citation searches

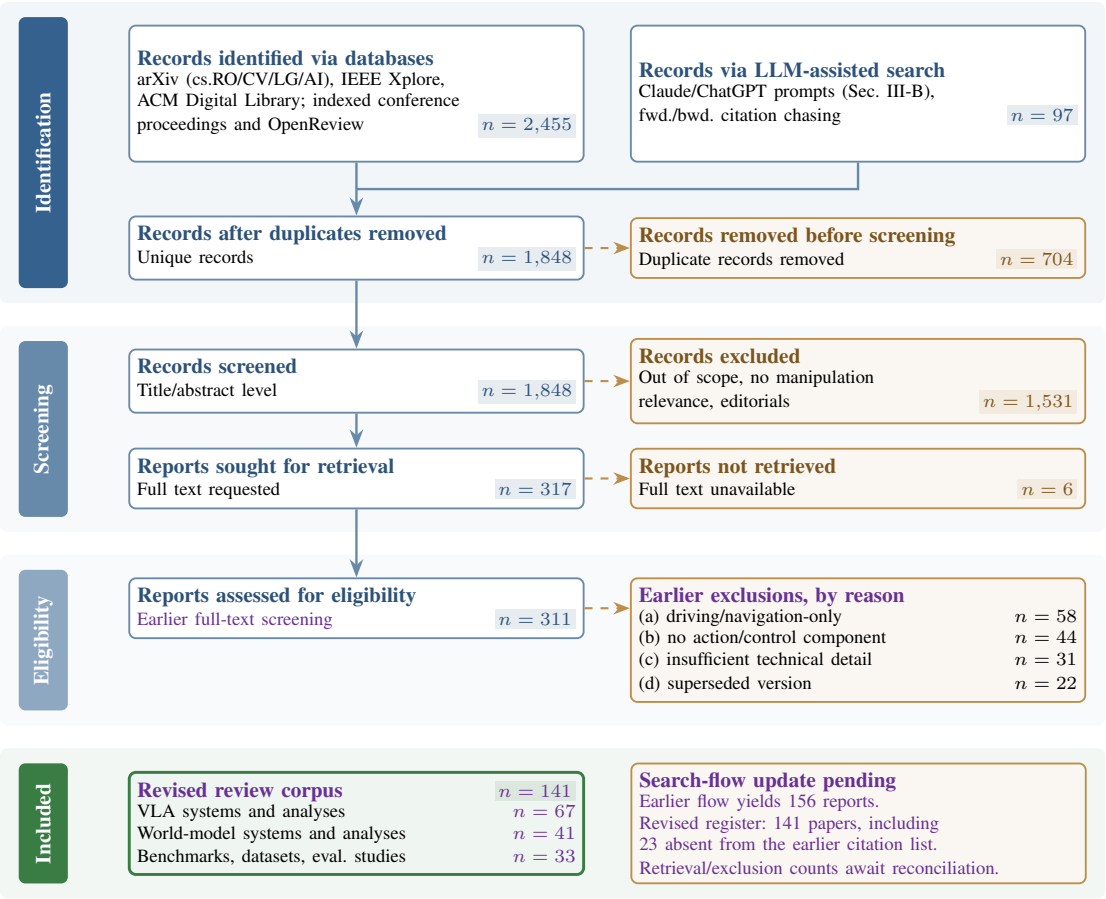


FIGURE 3. Reported identification and screening counts from the earlier August 2026 search, alongside the revised literature-register corpus updated through 1 October 2026. Blue and amber boxes retain the earlier reported search and exclusion counts. The separate green panel records 141 papers: 67 VLA, 41 world-model, and 33 benchmark, dataset, or evaluation papers. The updated register includes 23 papers absent from the earlier citation list. Its relationship to the historical screening counts has not yet been reconciled; no replacement retrieval or exclusion counts are inferred.

were performed from included surveys and influential systems.

Web-enabled ChatGPT and Claude were also used to support literature discovery. The same prompts were submitted to both systems. Representative prompts included: "List vision-language-action models for robotic manipulation published between January 2023 and August 2026." "Which world models have been applied to robotic manipulation or embodied control?" "Find studies that combine a world model and a VLA policy within one architecture." "Identify benchmarks for language-conditioned manipulation and robustness evaluation." "Find studies that use world models for reinforcement-learning post-training or synthetic robot data generation." "List surveys of VLA models, world models, and world-action models for robotics."

Language-model suggestions were used only for record identification. Each suggested publication was verified through an arXiv record or publisher page before screening. Unverifiable suggestions were discarded, and generated summaries were never used as evidence. Nine included reports were identified only through this discovery route. Duplicate records were removed using DOI and arXiv identifiers, followed by normalized title and author matching. Superseded versions were consolidated.

The search was designed primarily to identify systems and evaluation resources. The analysis of VLA limitations therefore represents a structured synthesis within the mapped corpus rather than an exhaustive estimate of all reported failures.

C. ELIGIBILITY CRITERIA

The inclusion criterion was a single retrievable paper, preprint, or technical report. Reports were eligible if they presented a VLA system, a world model, a training method, a dataset, a benchmark, or an evaluation study relevant to robotic manipulation. Peer-reviewed studies and technically detailed preprints were included. Reports on closed industrial systems were also retained because of their influence on the field, but their evidence was identified as vendor-reported.

Reports were excluded when they focused only on autonomous driving or navigation, contained no action or control component, or lacked sufficient technical detail for extraction. When several versions described the same work, the most complete version was retained. Related versions

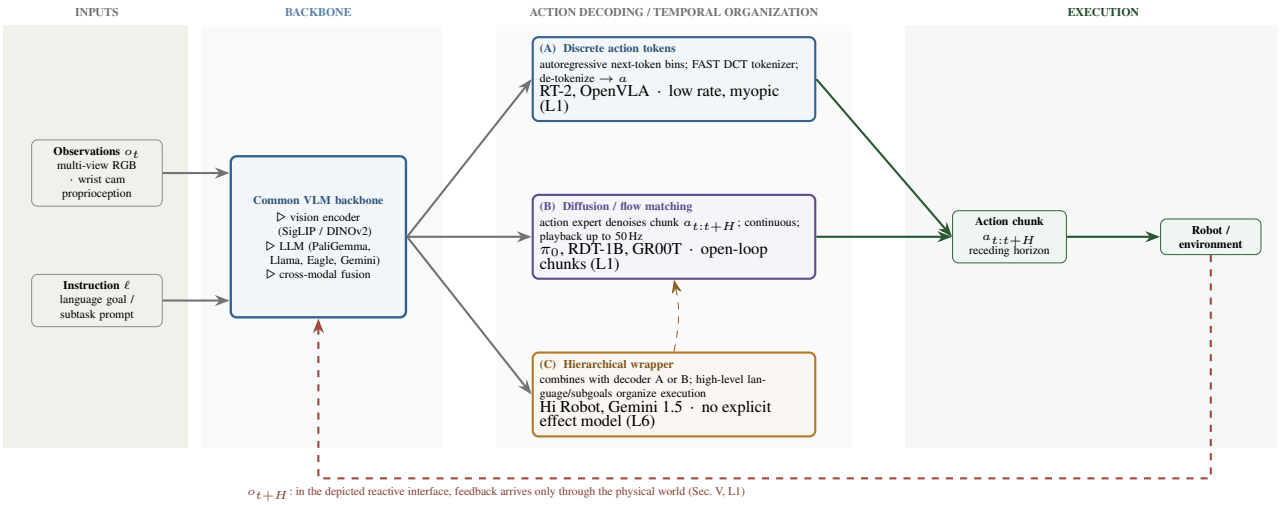


FIGURE 4. A common, but not universal, VLA architecture. Multi-view observations and a language instruction enter a VLM backbone (vision encoder, large language model (LLM), and cross-modal fusion), which feeds one of two action-decoder families: discrete tokens (A) or diffusion/flow matching (B). A hierarchical dual-system wrapper (C) is an orthogonal temporal-organization pattern that can use either decoder. Each is annotated with representative systems and the limitation it inherits. Execution is a receding-horizon action chunk; the dashed red loop marks the depicted reactive deployment interface, in which the controller does not expose an explicit action-conditioned consequence model for candidate comparison.

were linked through a common system identifier.

IV. VISION-LANGUAGE-ACTION MODELS

Vision-language-action (VLA) models connect visual perception and language understanding with robot control. Most systems adapt a pretrained vision-language backbone to predict robot actions, although the specific representation and training strategy vary substantially. Fig. 4 illustrates a common architecture in which visual observations and instructions are processed by a shared backbone before being passed to an action decoder.

The field cannot be described through a single architectural hierarchy. Some methods differ in how actions are represented, while others modify the training process or target a particular embodiment. We therefore organize the literature into seven overlapping groups according to the primary manipulation challenge addressed by each line of work. Fig. 5 summarizes this organization, and Table 2 provides a broader comparison. The groups are interpretive categories rather than mutually exclusive classes.

A. GROUP A: ACTION TOKENS

Early VLA research focused on transferring semantic knowledge from vision-language models to robotic control. RT-1 established a transformer policy with discretized action bins [38]. RT-2 extended this approach by representing robot actions as text tokens and jointly training on web-scale visual-language data and robot trajectories [1]. This formulation allowed knowledge of objects and semantic concepts to influence action prediction.

The Open X-Embodiment initiative expanded this idea across multiple robotic platforms [3]. OpenVLA subsequently provided an open implementation based on fused visual encoders and a large language backbone [2]. Related systems explored alternative multimodal prompting and lighter adaptation strategies [39]–[41]. LAPA further reduced dependence on labeled robot actions by learning latent action representations from video [42].

Later work addressed limitations of sequential token decoding. FAST introduced a frequency-space tokenizer for efficient action-sequence representation [43]. OpenVLA-OFT combined parallel decoding with continuous action prediction [8], while discrete-diffusion methods used iterative refinement to accelerate token generation [44]. These developments improve control resolution and efficiency, but the underlying policy still predicts actions directly. It does not explicitly model how alternative actions would change the environment.

B. GROUP B: CONTINUOUS ACTIONS

Continuous-action models address the precision and control-rate limitations of discrete tokenization. They commonly use diffusion or flow matching to generate smooth action sequences. This approach is particularly suitable for dexterous and bimanual manipulation, where several valid trajectories may satisfy the same instruction.

Octo demonstrated an open generalist policy with a diffusion decoder trained on Open X-Embodiment data [45]. The π_0 architecture paired a vision-language backbone with a flow-matching action expert [4]. RDT-1B used a diffusion transformer and a unified action space to support heterogeneous embodiments [46]. CogACT and DexVLA similarly separated semantic reasoning from continuous control through specialized action modules [47], [48]. These designs build on action chunking and diffusion-based control developed in earlier imitation-learning systems [49]–[51].

The same pattern appears in larger robotic foundation models. GR00T N1 combines a vision-language module

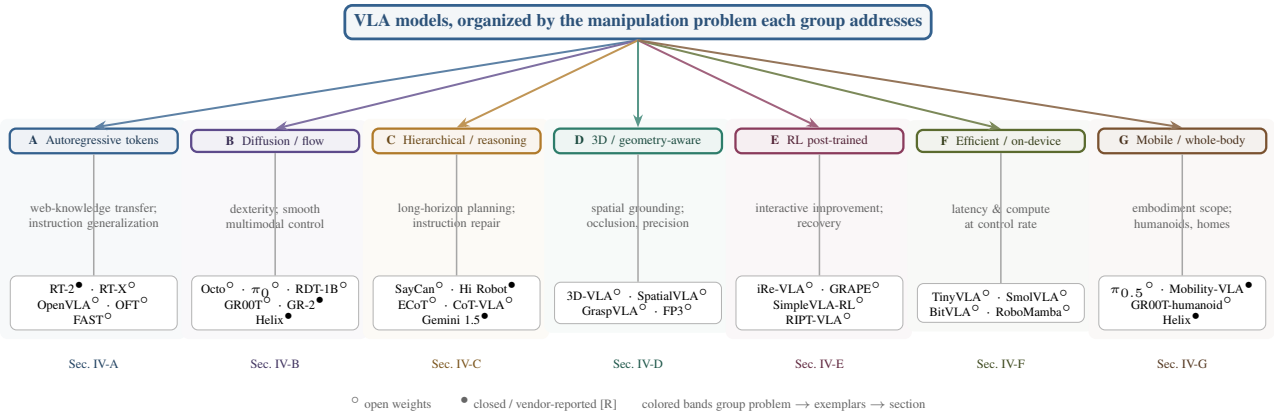


FIGURE 5. Problem-oriented taxonomy of VLA models (Sec. IV). Each of the seven groups is defined by the manipulation problem it addresses (italic sub-line), with representative systems below and the corresponding subsection; markers distinguish open-weight from closed, vendor-reported systems.

with a diffusion action head for humanoid control [5]. The $\pi_{0.5}$ model extends continuous action generation to mobile manipulation and long-duration tasks [52]. Closed systems such as Gemini Robotics and Figure Helix follow related dual-system designs but remain vendor-reported [6], [53].

Action chunking reduces repeated policy computation, but longer chunks also delay feedback. Reported control frequencies usually describe action playback rather than the rate at which the model generates a new plan. Continuous generation therefore improves motor smoothness without resolving the need for explicit consequence prediction or responsive replanning.

C. GROUP C: HIERARCHICAL REASONING

Long-horizon manipulation requires more than accurate local control. A robot must identify intermediate goals and recover when execution departs from the intended plan. Hierarchical methods address this problem by separating high-level deliberation from low-level action generation.

Early systems established several forms of this separation. SayCan grounded language-model plans using learned value functions [54]. PaLM-E incorporated continuous observations into an embodied language model [55], while RT-H introduced a language-motion hierarchy between instructions and actions [56]. Code as Policies and VoxPoser used generated programs or spatial value maps as interfaces between reasoning and control [57], [58].

More recent VLAs integrate reasoning more directly with action prediction. Hi Robot places an open-vocabulary policy above a continuous executor [59]. ECoT generates intermediate reasoning before actions [60], while OneTwoVLA switches between reasoning and execution according to the task context [61]. RationalVLA introduces feasibility reasoning for rejecting unsuitable instructions [62]. ReKep and MOKA use visual keypoints or marked images to connect semantic reasoning with geometric control [63], [64].

CoT-VLA is especially relevant to the predictive direction of this review because its reasoning process includes predicted future frames [65]. However, a single anticipated future does not provide systematic comparison among alternative actions. More broadly, learned hierarchical policies do not normally offer the feasibility guarantees or backtracking mechanisms associated with task-and-motion planning [66]. Their main contribution is improved task organization, while verified consequence reasoning remains open.

D. GROUP D: SPATIAL GROUNDING

VLA models built mainly on two-dimensional images can struggle with metric placement and viewpoint changes. Spatially grounded methods address this limitation by incorporating depth, point clouds, voxels, or learned three-dimensional features.

PerAct and RVT established the value of voxel and multi-view representations for precise manipulation [67], [68]. Later systems improved the efficiency and accuracy of these representations [69], [70]. Within the VLA paradigm, 3D-VLA connected a three-dimensional language model with diffusion-based action generation [71]. SpatialVLA encoded egocentric geometry as spatial action tokens [72], while PointVLA incorporated point clouds into a pretrained VLA [73]. Lift3D and FP3 transferred foundation-model features into three-dimensional policy representations [74], [75].

Other systems combine semantic priors with task-specific geometric interfaces. BridgeVLA aligns vision-language features with spatial heatmaps [76], and GraspVLA uses large-scale synthetic grasp data to support spatially grounded grasping [77]. GeoVLA and 3D-CAVLA further integrate geometric encoders with contextual reasoning [78], [79].

These methods improve knowledge of where objects and targets are located. This is not equivalent to learning how objects will move or deform under contact. Spatial grounding strengthens the state representation, whereas consequence prediction requires a model of dynamics.

E. GROUP E: INTERACTIVE LEARNING

Supervised VLA policies inherit the coverage of their demonstrations. When execution moves outside that cover-

age, the policy may lack corrective or recovery behavior. Reinforcement-learning post-training addresses this limitation through interaction and task-level feedback.

Several methods combine reinforcement learning with supervised stabilization. iRe-VLA alternates online improvement with supervised consolidation [80]. GRAPE uses trajectory-level preferences [81], while RLDG transfers skills from specialist reinforcement-learning policies into a generalist VLA [82]. ConRFT applies reinforced fine-tuning on physical robots [83], and VLA-RL scales policy optimization using learned progress signals [84].

Recent studies report substantial gains in particular LIBERO configurations. SimpleVLA-RL improves an OpenVLA-OFT starting point through online interaction [11], while RIPT-VLA uses sparse rewards to improve a weak supervised policy [12]. FLaRe also reports generalization gains from large-scale reinforcement-learning adaptation [85]. These results show that supervised policies can be improved through interaction. They do not establish whether the benefit comes from reinforcement learning alone because the data distribution and feedback signal also change.

The practical limitation is that physical interaction is expensive and may be unsafe. Conventional simulators provide one solution, while learned world models offer another. The latter could extend interactive training beyond available demonstrations, but their value depends on whether improvements obtained in the learned environment transfer to the real robot.

F. GROUP F: EFFICIENT DEPLOYMENT

On-robot deployment imposes strict limits on memory and inference time. Efficiency research reduces these costs through smaller backbones, alternative sequence models, compressed weights, and faster decoding.

TinyVLA and SmolVLA show that compact policies can retain competitive task performance [86], [87]. EdgeVLA further adapts non-autoregressive action prediction to embedded hardware [88]. RoboMamba replaces standard attention with state-space dynamics [89], while BitVLA combines low-bit weights with vision compression [90]. Other methods exploit temporal redundancy or prune unnecessary computation [91], [92].

Decoding can also be adapted to the difficulty of the current situation. PD-VLA and CEED-VLA accelerate action generation through parallel prediction or early exit [93], [94]. DeeR-VLA applies adaptive exits within the multimodal backbone [95]. Gemini Robotics On-Device represents a closed implementation of the same deployment objective [96].

Efficiency is not only a hardware concern. Reducing policy cost can support more frequent replanning or release computation for uncertainty estimation and predictive search. However, a faster policy is not necessarily more reliable. The key question is whether saved computation can be converted into better closed-loop reasoning within the available control deadline.

G. GROUP G: EMBODIMENT

Moving beyond tabletop manipulation introduces mobile bases, whole-body motion, and humanoid platforms. These embodiments increase the state dimension and amplify the consequences of planning errors.

OWMM-Agent combines scene understanding with navigation and grasping in a mobile manipulation system [97]. OK-Robot examines how open components can support manipulation in unfamiliar homes [98]. The $\pi_{0.5}$ model extends a generalist policy to mobile manipulation over long task sequences [52], while Long-VLA and LoHoVLA address error accumulation across extended execution horizons [99], [100].

Humanoid systems further require coordination across the upper body or the entire platform. GR00T N1 and N1.5 provide open foundation models for humanoid control [5], [101]. Figure Helix demonstrates high-rate upper-body control on commercial hardware, although its results remain closed and vendor-reported [53]. These systems broaden the scope of VLA policies, but evidence across platforms remains difficult to compare because embodiments and evaluation protocols differ substantially.

As the platform becomes more capable, the cost of an incorrect action also increases. Embodiment scaling therefore strengthens the case for predictive evaluation and runtime assurance rather than serving only as another measure of policy generality.

Summary. The seven groups capture complementary responses to the central challenges of VLA-based manipulation. Action-token and continuous-action models improve semantic transfer and motor generation. Hierarchical and spatially grounded systems extend reasoning and representation. Interactive learning addresses weaknesses inherited from supervised data, while efficiency and embodiment research target deployment constraints. These advances increasingly blur the boundaries between categories, but most systems still select actions without explicitly comparing their predicted consequences. The next section examines the limitations that follow from this gap and from current evaluation practice.

V. LIMITS OF CURRENT VLA SYSTEMS

Strong benchmark performance does not establish that a VLA system can predict consequences, remain reliable under distribution shift, or operate safely in the physical world. This section examines seven limitations that recur across the literature. They arise at different levels of the system and should not be interpreted as properties of every VLA model. Predictive models may address some of these limitations, but they are not universal solutions. Table 3 therefore distinguishes evidence that a limitation exists from evidence that a world-model mechanism improves it.

A. L1: CONSEQUENCE AWARENESS

Most VLA policies map observations and instructions directly to an action or action sequence. They may encode temporal information internally, but they do not normally expose

TABLE 2. Representative vision-language-conditioned manipulation policies and VLA-based systems by problem-oriented group. Standalone language planners, embodied VLMs without an action policy, and navigation-only systems are excluded. Groups may overlap; year denotes the first public report. Action: AR = autoregressive, Diff = diffusion, FM = flow matching, Hier = hierarchical. Open: ✓ = official public code or weights, (✓) = only part of the model family or artifacts released, — = no verified release. Tags: [T] primary-source traced, [R] vendor-reported, [P] protocol-unresolved (Sec. VII-B). †Pre-2023 language-conditioned action-policy precursor. Headline results use each source's protocol and are not comparable across rows.

Model	Year	Grp	Backbone	Action head	Open	Tag	Headline (source protocol)
RT-1† [38]	2022	A	EfficientNet + Transformer	Parallel action bins	✓	[T]	Language-conditioned control; 130k episodes
RT-2 [1]	2023	A	PaLI-X/PaLM-E (55B)	AR text tokens	—	[R]	Emergent semantic generalization
RT-X [3]	2023	A	RT-1 / RT-2	Discrete / AR tokens	(✓)	[T]	Cross-embodiment transfer; RT-1-X checkpoint released
OpenVLA [2]	2024	A	Llama-2-7B	AR bins	✓	[T]	+16.5 percentage points over RT-2-X (55B)
FAST / π_0-FAST [43]	2025	A	PaliGemma-3B	AR (DCT tokens)	✓	[T]	AR at dexterous control rates
OpenVLA-OFT [8]	2025	A	Llama-2-7B	Parallel cont. (ℓ_1 loss)	✓	[T]	LIBERO 95.3% or 97.1%, config.-specific
Octo [45]	2024	B	Transformer (27M/93M)	Diff chunks	✓	[T]	800k OXE episodes; language/image goals
π_0 [4]	2024	B	PaliGemma-3B+300M	FM chunks; 50-Hz execution	✓	[T]	10k+ h teleop, 7 platforms
RDT-1B [46]	2024	B	1.2B DiT	Diff chunks	✓	[T]	1M+ pretraining episodes; 6k+ ALOHA fine-tuning episodes
GR00T N1 [5]	2025	B	Eagle-2 (1.34B)	DiT FM, dual-sys	✓	[T]	16-act chunk / 63.9 ms (L40)
GR00T N1.7 [102]	2026	B/G	Cosmos-Reason2-2B	DiT FM chunks	✓	[T]	Open 3B cross-embodiment policy
$\pi_{0.5}$ [52]	2025	B/G	PaliGemma-class 4B	FM + high-level	✓	[T]	Unseen-home mobile cleaning
$\pi_{0.7}$ [103]	2026	B/C	Gemma3-4B + 860M expert	FM + multimodal prompts	—	[R]	Language/subgoal steering; cross-embodiment transfer
LingBot-VLA 2.0 [104]	2026	B/D/G	Qwen3-VL-4B + MoE	FM; 55-D actions	✓	[T]	Whole-body actions; future-feature supervision
GR-2 [105]	2024	B	Video transformer (230M)	cVAE action chunks	—	[R]	97.7% over 105 internal tabletop tasks
Gemini Robotics [6]	2025	B	Gemini 2.0	Closed	—	[R]	Frontier dexterity (vendor)
Helix [53]	2025	B/G	7B S2 + 80M S1	Dual-sys, 200 Hz	—	[R]	Whole-upper-body humanoid
Helix 02 [106]	2026	B/G	S2 + S1 + learned S0	Joint targets + whole-body ctrl.	—	[R]	Continuous loco-manipulation with tactile input
Hi Robot [59]	2025	C	PaliGemma-3B + π_0	Hier + FM chunks	—	[T]	Hierarchical reasoning with a learned VLA executor
ECoT [60]	2024	C	OpenVLA-base	AR + reasoning	✓	[T]	+CoT before acting
CoT-VLA [65]	2025	C	VILA-U (7B)	AR frames + action chunks	—	[T]	Predicted visual subgoals guide closed-loop actions
LoHoVLA [100]	2025	C	PaliGemma-3B	AR language + actions	—	[T]	20 long-horizon Ravens tabletop tasks
Gemini Robotics 1.5 [7]	2025	C	Gemini + ER 1.5	Hier, thinking	—	[R]	Interleaved reasoning, motion transfer
3D-VLA [71]	2024	D	3D-LLM	Diff (3D)	✓	[T]	3D world-model-style heads
SpatialVLA [72]	2025	D	PaliGemma2-3B	Spatial AR tokens	✓	[T]	4B policy; 1.1M real-robot episodes
GraspVLA [77]	2025	D	InternLM2 + vision	AR perception + FM	✓	[T]	Billion-frame synthetic grasp data; open-vocabulary grasping
FP3 [75]	2025	D	Uni3D + CLIP + DiT	Diff chunks	✓	[T]	Language-conditioned 3D policy; 60k DROID episodes
SimpleVLA-RL [11]	2025	E	Modified OpenVLA-OFT	Parallel action tokens	✓	[T]	Online RL; LIBERO and RoboTwin evaluations
RIPT-VLA [12]	2025	E	QueST / OpenVLA-OFT	AR / continuous chunks	✓	[T]	RL post-training with sparse success rewards
FLaRe [85]	2024	E	SPOC transformer	Discrete actions	✓	[T]	PPO tuning; 79.5% CHORES-S (simulated navigation/manipulation)
TinyVLA [86]	2024	F	Compact VLM family	Diff	✓	[T]	0.4–1.3B family; no robot pretraining
SmolVLA [87]	2025	F	SmolVLM2 + expert	FM	✓	[T]	450M policy; consumer-GPU training
BitVLA [90]	2025	F	BitNet-2B + SigLIP	Parallel cont. MLP	✓	[T]	Ternary-weight backbone; OFT-style action regression
RoboMamba [89]	2024	F	CLIP + Mamba-2.7B	MLP pose regression	(✓)	[T]	Efficient pose prediction; training code by request

an action-conditioned model for comparing the consequences of alternative actions. This distinction matters because errors can move the robot beyond the distribution represented by its demonstrations [107].

Predictive supervision provides a partial response. World-VLA jointly learns action generation and future-frame prediction [16], but its reported gains do not isolate the value of explicit candidate comparison. Cosmos Policy evaluates a package that refines rollouts and selects among several candidates [17]. The evidence is direct but limited to two ALOHA tasks and does not isolate search from the other components. The absence of an explicit consequence interface is therefore clear, while its general effect on task performance remains insufficiently measured.

B. L2: DISTRIBUTION SHIFT

Behavior cloning learns from the states and actions represented in its demonstrations. Performance may deteriorate when execution reaches unfamiliar states or requires recovery behavior that was not observed during training. The severity of this problem depends on data coverage and the policy's inductive structure.

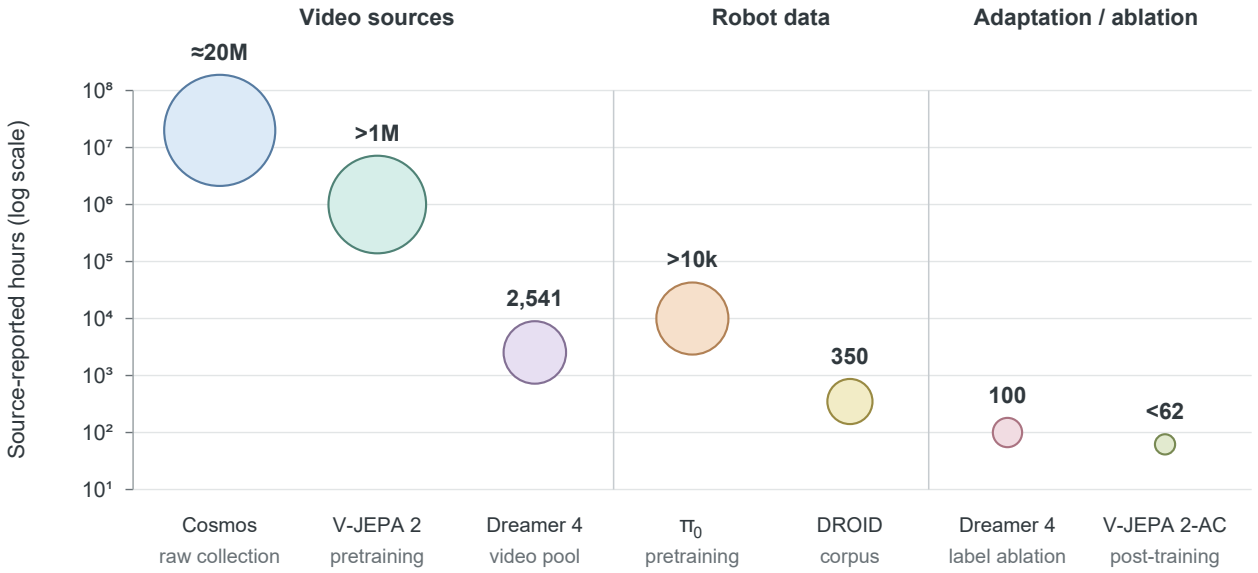


FIGURE 6. Source-reported data scales shown as light-filled bubbles on a logarithmic hours axis [4], [13], [15], [108], [109]. Colors distinguish the plotted records. Bubble area increases linearly with the logarithm of the reported value or bound threshold; areas are not proportional to raw hours. The numerical labels retain approximate values and lower or upper bounds. Cosmos denotes the raw candidate collection, π_0 a robot pretraining mixture, and DROID its reported demonstration corpus. Dreamer 4's 100-hour action-label generation ablation is nested within its 2,541-hour gameplay pool (90/10 training/evaluation split), not the full agent-training budget. V-JEPA 2-AC denotes robot-video post-training with states and derived actions, excluding earlier video pretraining. Different tasks, supervision, and populations prevent a sample-efficiency ranking.

Reinforcement-learning post-training can substantially improve particular supervised policies. SimpleVLA-RL, RIPT-VLA, and FLaRe report large gains after introducing interaction and reward feedback [11], [12], [85]. However, these studies change the data distribution and optimization objective together. They therefore demonstrate that the original policies can be improved, but do not establish a universal performance ceiling for imitation learning. Dataset aggregation and interactive learning remain valid alternatives to world-model-based practice.

C. L3: DATA REQUIREMENTS

VLA models inherit broad visual and semantic knowledge from pretrained backbones, but grounding that knowledge in robot actions still requires expensive interaction data. The scale of current efforts illustrates this burden. The π_0 dataset contains more than 10,000 hours of teleoperation [4], DROID contributes 350 hours of robot interaction [108], and Open X-Embodiment aggregates 970,000 trajectories across many institutions [3].

Predictive systems can exploit larger collections of passive video. V-JEPA 2 uses extensive action-free video before action-conditioned adaptation [13], while Dreamer 4 separates passive gameplay data from a smaller paired-action subset [15]. These examples show that passive data can support predictive representation learning. They do not quantify how many hours of robot interaction are replaced because the tasks and supervision differ. Fig. 6 therefore separates passive video from action-conditioned and robot-demonstration data.

D. L4: PHYSICAL GROUNDING

Visual plausibility does not guarantee correct physical behavior. A model may generate realistic images while misrepresenting contact or object dynamics. Geometric accuracy must also be distinguished from causal validity because knowing where an object is does not establish how it will respond to an action.

Physics-IQ reports a substantial gap between generated and real-video performance and finds no statistically significant positive association between its realism and physics scores [110]. WorldModelBench and VBench-2.0 similarly identify physical inconsistencies in video generation [111], [112]. These benchmarks concern video models rather than deployed VLA policies, so they provide indirect evidence. Direct evaluation should vary physical properties while controlling visual appearance and then measure both prediction error and policy behavior. The main finding is therefore an evaluation gap rather than proof that current VLAs lack physical understanding.

E. L5: EVALUATION VALIDITY

Several VLA systems report near-saturated performance on standard LIBERO configurations [8], [11], [17]. However, LIBERO-Plus shows that performance can decline sharply under changes in viewpoint, layout, or language [9]. This contrast indicates that success on the original benchmark may not reflect robust instruction grounding or environmental generalization.

Simulation introduces another layer of uncertainty. PolaRiS shows that generated rollouts may contain hallucinated

interactions and may not preserve real-world policy rankings [10]. SimplerEnv was developed to improve correspondence between simulation and physical evaluation [113], but simulator performance still depends on the selected tasks and calibration procedure. Benchmark results should therefore be interpreted within their original protocols. A learned evaluator or larger simulator does not automatically provide valid real-world comparison.

F. L6: SAFETY AND UNCERTAINTY

Standard behavior-cloning objectives do not require calibrated uncertainty or estimates of constraint violation. They also do not identify when an action may cause irreversible harm. These capabilities must be introduced through the learning objective or through a separate assurance layer.

LIBERO-Safety evaluates policies in safety-critical manipulation settings [114], while ASIMOV accompanies the vendor-reported safety framework of Gemini Robotics [7]. Other approaches use uncertainty to request assistance or abstain from unsupported actions [115]. Control barrier functions and reachability methods can provide runtime protection without a generative world model [116], [117].

A world model may make predicted outcomes available for inspection, but prediction is neither necessary nor sufficient for safety. An inaccurate model can produce confident and unsafe forecasts. Consequence prediction must therefore be combined with uncertainty calibration, runtime enforcement, and a tested fallback policy.

G. L7: DELIBERATION COST

Prediction and search compete with the timing requirements of robot control. Generating future trajectories or evaluating many candidate actions may improve deliberation but also delay the next policy update. Dual-system architectures address this tension by separating fast execution from slower reasoning [5], [53].

The V-JEPA 2 study illustrates the scale of the challenge. Its latent planner is reported to operate much faster than a pixel-space generative baseline, but the two configurations use different models and candidate counts [13]. The result supports the efficiency of latent prediction in that setting but does not provide a controlled comparison. Planning latency must also be distinguished from servo rate and action playback. The practical question is not whether prediction is computationally expensive in isolation, but whether it can improve decisions within the available control deadline.

Summary. The seven limitations arise from different parts of the VLA pipeline. Some concern policy learning and data coverage, while others reflect weaknesses in evaluation or system assurance. The least developed evidence concerns explicit candidate-action comparison and consequence-aware safety. The next section examines how world models could address these gaps and where their benefits have been demonstrated.

VI. WORLD MODELS AND PREDICTIVE MECHANISMS

World models extend robot learning beyond direct action prediction by learning how the environment changes over time. Their role varies across systems. Some support planning in a compact latent space, while others generate future observations or provide imagined experience for policy learning. More recent architectures combine prediction and action generation within a shared model (Fig. 12).

A. SCOPE AND TAXONOMY

Under partial observability, a world model infers a state representation from the available observation and action history. The state is then propagated through an action-conditioned transition model:

$$z_t \sim q_\phi(z_t \mid h_t), \qquad p_\theta(z_{t+1} \mid z_t, a_t). \tag{1}$$

The predicted state may be decoded into future observations or used to estimate task value and termination. Language can also condition the transition when the instruction changes which aspects of the environment are relevant.

This definition concerns model structure rather than downstream use. The same predictive model may support planning, imagined policy learning, synthetic data generation, or policy evaluation. We also include predictive components learned from passive video when an action-conditioned adaptation is provided. However, action conditioning alone does not establish causal validity. Reliable candidate comparison also requires sufficient state information and support for the evaluated actions.

The literature is organized into six overlapping families according to prediction space and functional role. The comparison in Fig. 7 distinguishes direct policies from planning through a model and from unified world-action architectures. Fig. 8 summarizes the six families, while Table 4 provides representative systems. These categories are not mutually exclusive because a single model may serve several predictive functions.

B. WM1: LATENT DYNAMICS

Latent-dynamics models learn a compact state in which future outcomes can be predicted without generating every image detail. This representation supports efficient planning and policy learning through imagined trajectories.

The recurrent World Models framework established the idea of training a policy inside a learned environment [118]. PlaNet later introduced practical planning with a recurrent state-space model [119]. The Dreamer family replaced online search with actor-critic learning in imagination and demonstrated strong performance across diverse control domains [120]. MuZero showed that a useful model need not reconstruct observations when it can predict quantities that matter for decisions [121]. TD-MPC2 further scaled latent planning across continuous-control tasks [122].

Dreamer 4 provides an important recent example of learning entirely inside a world model from offline data [15]. Its Minecraft result demonstrates that imagined learning can

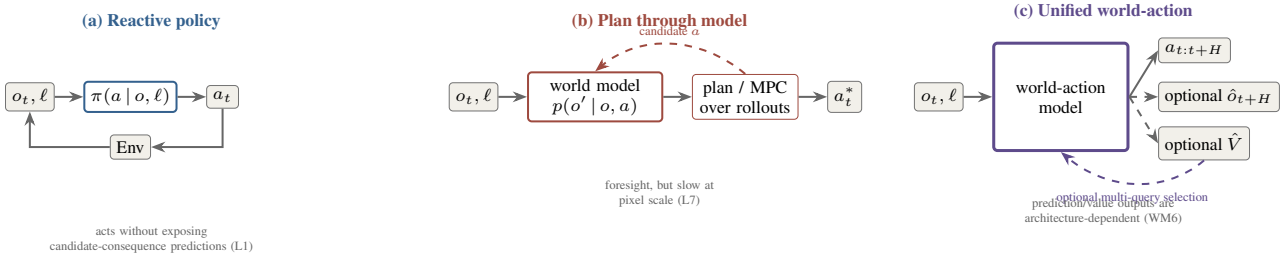


FIGURE 7. Three control interfaces. (a) A policy maps observations directly to actions without exposing a consequence model. (b) An explicit predictive model permits candidate search, with cost determined by representation and search. (c) A unified model shares representations for predictive and action objectives; future-state and value outputs are architecture-dependent and need not be emitted during every direct-policy inference. Explicit best-of-N selection still requires repeated queries and is not near-policy latency by default.

support complex behavior with limited action-labeled data. However, it remains a cross-domain proof of concept rather than direct evidence for general robotic manipulation. Latent dynamics offer a promising response to interaction cost and planning latency, but their benefit must still be established under physical contact and real-world distribution shift.

C. WM2: GENERATIVE VIDEO MODELS

Generative world models predict future observations in pixel space. Their main advantage is the ability to learn broad visual dynamics from large video collections and present predicted outcomes in an interpretable form.

UniPi formulated decision making as language-conditioned video generation followed by inverse dynamics [123]. UniSim extended video prediction toward interactive learned simulation [124]. Related work extracts actions from passive video or generates intermediate visual goals for control [125]–[128]. GR-1 and GR-2 brought video-predictive pretraining into manipulation policies [105], [129].

Large world foundation models extend this approach beyond individual tasks. Cosmos provides an open platform for video tokenization and generative dynamics [109]. The Genie family focuses on interactive environments that respond to user control [130], [131]. Genie Envisioner adapts this idea to manipulation by combining video prediction, action generation, and learned simulation [132].

These models can provide transferable visual priors and scalable imagined environments. Their central limitation is that visual plausibility does not guarantee physically correct dynamics. Pixel generation is also computationally demanding when many candidate actions must be evaluated. Generative video models therefore provide an expressive predictive interface, but their value for control depends on physical validity and efficient inference.

D. WM3: LATENT PREDICTION

Latent-predictive models avoid reconstructing pixels and instead forecast task-relevant representations. The goal is to retain useful dynamics while reducing the cost and uncertainty associated with detailed visual generation.

I-JEPA and V-JEPA established representation-space prediction for images and video [133], [134]. V-JEPA 2 scales this objective through action-free video pretraining, while V-JEPA 2-AC introduces an action-conditioned predictor for robotic planning [13]. The adapted model outperforms Octo on two reported pick-and-place tasks without task-specific demonstrations. Its planning time is also substantially lower than that of the evaluated pixel-space baseline. However, the comparison uses different models and candidate counts, and the reported latency remains above typical control rates.

DINO-WM demonstrates that pretrained visual features can support planning without pixel reconstruction [135]. LaWAM predicts latent visual subgoals for a VLA policy [136], while VLA-JEPA integrates latent prediction more directly with action learning [137]. These systems suggest that latent prediction may provide a better balance between foresight and inference cost. Evidence for the contribution of passive video remains indirect because current studies do not isolate it through controlled data ablations.

E. WM4: PHYSICAL STATE MODELS

Pixel and latent representations may omit persistent geometry or contact state. Physical world models instead predict within representations that make three-dimensional structure and interaction more explicit.

TesserAct predicts multiple visual and geometric modalities to construct temporally consistent scenes [138]. Gaussian World Models propagate dynamics through explicit Gaussian representations [139], while ManiGaussian applies dynamic splatting to manipulation [140]. PhysDreamer estimates material properties for physically grounded interaction [141]. Tactile world-action models extend prediction to contact-rich behavior [142], [143].

This family addresses physical grounding more directly than appearance-based prediction. It may also provide state variables on which constraints can be imposed. However, comparative manipulation evidence remains limited. Explicit geometry improves the representation of the world, but it does not by itself guarantee accurate dynamics or safe control.

F. WM5: LEARNED ENVIRONMENTS

World models can also serve as environments in which data are generated, policies are evaluated, or reinforcement-

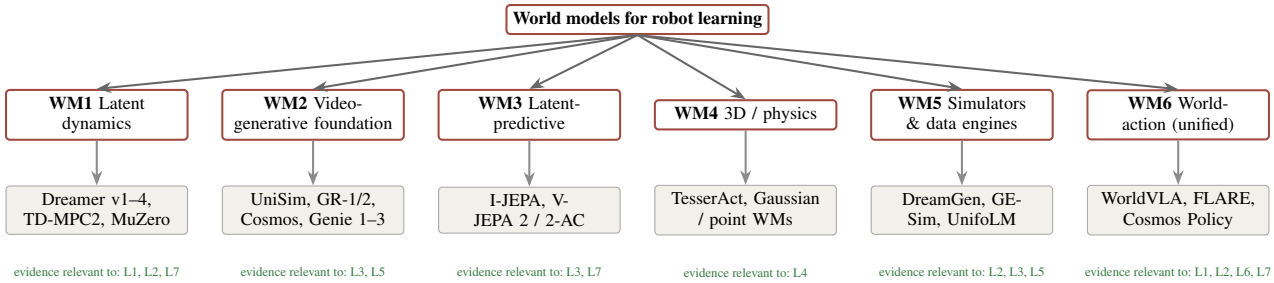


FIGURE 8. Overlapping functional families of predictive systems (Sec. VI), with representative systems and candidate mechanisms relevant to the VLA limitations (L1–L7, Sec. V).

learning agents are trained. These roles are attractive because physical interaction is expensive and difficult to scale.

As a data engine, DreamGen generates visual trajectories for new tasks and labels them through inverse dynamics [14]. Training with these trajectories improves policy performance on previously unsupported behaviors. MimicGen and DexMimicGen provide related precedents through simulator-based recombination of demonstrations [144], [145]. DreamGen extends the idea by using a learned video model rather than a conventional simulator.

As an evaluator, an action-conditioned model can predict the outcomes of a candidate policy without deploying it on hardware [146]. GE-Sim applies this idea within Genie Envisioner [132]. However, PolaRiS shows that generated evaluations may hallucinate interactions or rank policies differently from physical evaluation [10]. Learned simulation is therefore not automatically a valid evaluation instrument.

World-Gymnast provides direct evidence for training a policy inside a learned environment and transferring it to a real robot [147]. Its learned-world-model package outperforms a conventional simulator on three of four tasks but performs worse on the fourth (Fig. 9). The experiment demonstrates feasibility while also showing why task-specific outcomes should not be hidden by a single average. The comparison is narrow and does not isolate world-model quality from the rest of the training pipeline.

The strongest evidence for this family concerns synthetic augmentation and narrow learned-practice experiments. General policy evaluation and transferable imagined training remain unresolved.

G. WM6: WORLD-ACTION MODELS

World-action models combine predictive and control objectives within a shared architecture. This design can use prediction to improve the learned representation without requiring a separate planner during every action query. It can also expose future-state or value predictions when more deliberate decision making is needed.

WorldVLA jointly learns action generation and future-frame prediction [16]. Its ablations show that predictive supervision can improve policy performance, although the gains depend on the action-decoding configuration. The study does not evaluate explicit rollout-based comparison among alternative actions. Auxiliary prediction and candidate-action search should therefore be treated as different mechanisms.

FLARE aligns policy representations with predicted future latents and supports training with action-free video [148]. DreamVLA and tactile world-action systems extend joint prediction toward broader world knowledge and contact-rich interaction [142], [143], [149]. StructVLA uses sparse structured foresight [150], while LaWAM predicts latent subgoals to reduce inference cost [136]. These methods explore different positions on the trade-off between predictive detail and control latency.

Cosmos Policy adapts a video world model into a policy that predicts actions together with future states and values [17]. Its direct-policy mode and its planning package are separate evaluations. The planning package reports an improvement on two ALOHA tasks after rollout refinement and repeated candidate selection, but it combines several interventions and substantial computation. The result provides direct but narrow evidence for the complete package rather than for search alone.

Unified architectures show that prediction and action can reinforce one another within a shared model. They do not yet establish that every policy benefits from explicit planning or that predicted values provide calibrated safety information.

H. EVIDENCE MAP

Table 3 connects the seven VLA limitations to the predictive mechanisms discussed above (Fig. 11). The central distinction is between evidence that a limitation exists and evidence that a proposed mechanism changes the relevant outcome.

The clearest matched evidence concerns synthetic trajectory augmentation. Auxiliary predictive training also improves particular world-action policies. Evidence for learned practice is direct but restricted to the four World-Gymnast tasks. Candidate-action comparison is evaluated only through the bundled Cosmos Policy experiment. Physical grounding and passive-video attribution remain supported mainly through indirect evidence. Learned evaluation includes adverse findings from PolaRiS, while consequence-aware safety remains prospective.

The matrix should therefore be read as a map of testable relationships rather than a causal graph. An association be-

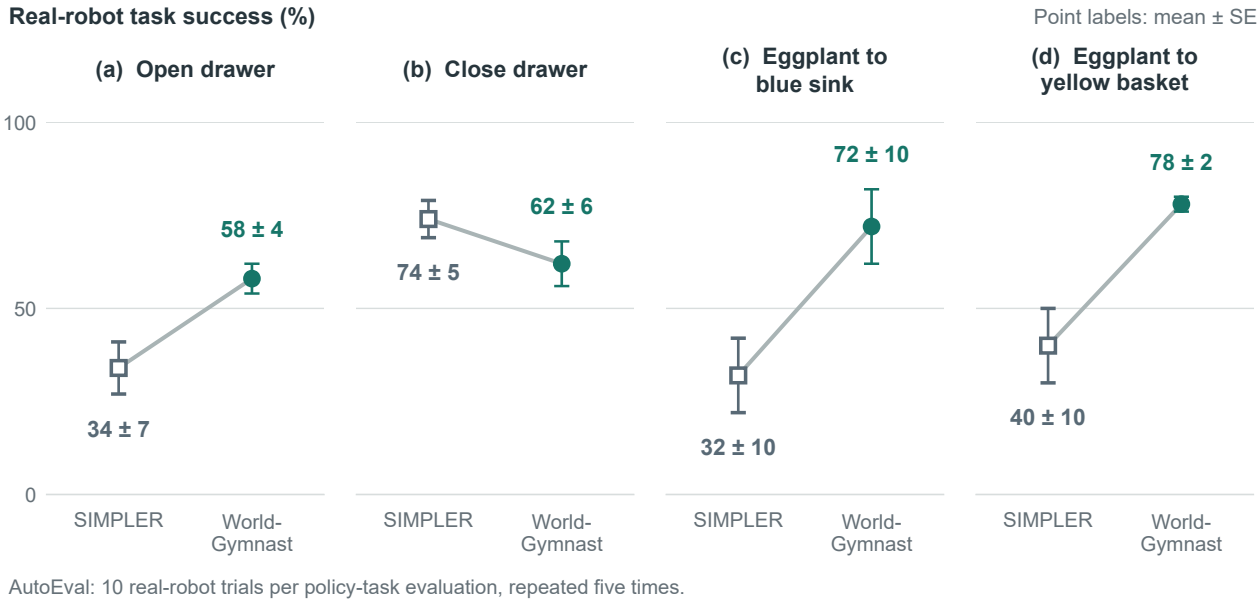


FIGURE 9. Task-level outcomes after reinforcement learning in SIMPLER and the World-Gymnast learned environment [147]. Squares and circles show source-reported means; vertical bars and numerical labels give the source-reported standard errors (SEs). Each policy–task evaluation used 10 real-robot trials and was repeated five times. All panels share a 0–100% scale. Connecting segments join the two method-level means; they do not represent paired trial results or a training trajectory. The four-task package-level comparison does not isolate the effect of world-model quality.

tween a limitation and a mechanism does not imply that the mechanism is necessary or sufficient.

I. LIMITATIONS OF WORLD MODELS

World models introduce their own sources of error. Perceptual quality does not guarantee control-consistent dynamics. Physics-IQ finds no statistically significant positive relationship between visual realism and its physics score across the evaluated models [110]. Long-horizon predictions can also drift as small model errors accumulate. Prediction moves the source of error from the policy to the learned environment rather than eliminating it.

Computational cost remains a second challenge. Pixel-space planning can require minutes for one action decision, while latent planning is faster but may still exceed the control deadline [13]. Unified architectures reduce repeated computation when prediction is used only as a training objective. Explicit comparison of many candidates still requires additional inference.

Learned simulators also inherit biases from their training data. PolaRiS demonstrates that generated rollouts can hallucinate interactions and mis-rank policies [10]. Synthetic trajectories may pass related errors into the policy that learns from them. These risks become harder to assess when the model or evaluation system is closed, as in several vendor-reported platforms.

World models should therefore be judged by decision utility and transfer to physical control rather than by prediction quality alone. Current evidence does not support a general conclusion that predictive systems outperform direct policies across robotic manipulation.

Summary. World models now support latent planning, generative simulation, representation learning, physical-state prediction, imagined practice, and unified action generation. The most convincing manipulation evidence concerns synthetic data and auxiliary predictive training. Evidence for candidate comparison and real-world learned practice remains narrow. Physical validity, evaluation transfer, control-rate planning, and consequence-aware safety remain open. The following section examines these claims under their reported evaluation protocols.

VII. EVALUATION EVIDENCE

Evaluation remains fragmented across VLA policies and world models. Policy benchmarks primarily measure task completion, generalization, and robustness, whereas world-model benchmarks emphasize perceptual quality, physical consistency, or predictive accuracy. Few protocols evaluate whether prediction improves manipulation decisions. This separation limits cross-family comparison and creates a central measurement gap: prediction quality, decision utility, and control performance are related but not interchangeable.

A. BENCHMARK SCOPE

VLA evaluation spans simulated manipulation, matched simulation and physical platforms, and distributed real-robot testing. LIBERO is widely used for language-conditioned tabletop manipulation [155], while LIBERO-Plus evaluates robustness under changes in viewpoint, layout, language, and other environmental factors [9]. CALVIN emphasizes

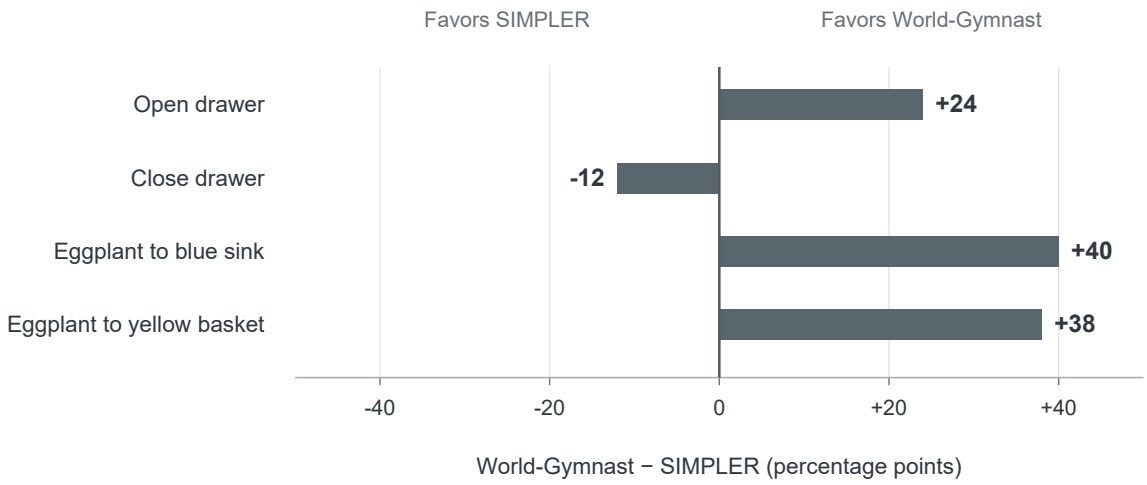


FIGURE 10. Differences in the source-reported real-robot task-success means from Fig. 9 [147]. Each bar shows World-Gymnast minus SIMPLER in percentage points (pp); positive values favor World-Gymnast and negative values favor SIMPLER. The axis is symmetric about zero. These four descriptive differences use the same AutoEval experiment, not additional evaluations. Source-reported standard errors and the evaluation protocol are retained in Fig. 9; no difference confidence interval or statistical significance is inferred.

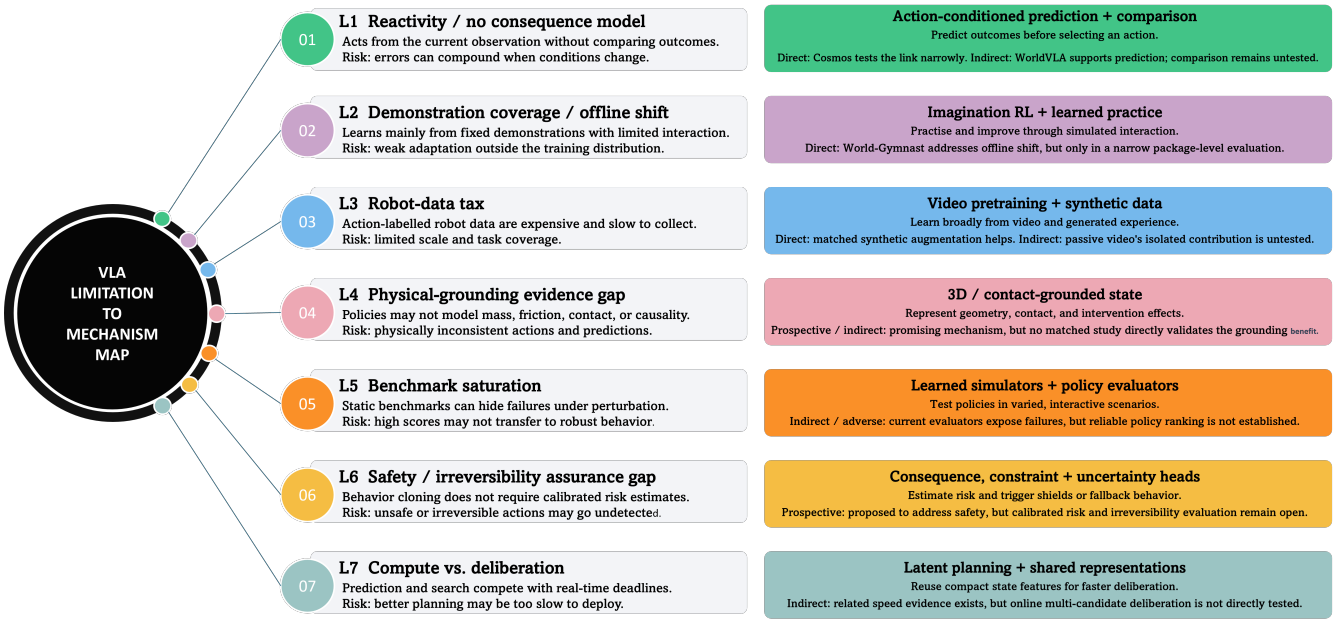


FIGURE 11. Limitation-to-candidate-mechanism map. Edge style summarizes the reported mechanism-effect evidence: solid direct, dashed indirect or adverse, and dotted prospective. The map is not a causal graph, and an edge does not imply that the mechanism is necessary or sufficient. Consequence-aware safety (L6) remains the least directly evaluated connection.

long-horizon task sequences and generalization across environments [156]. RLBench, ManiSkill, RoboCasa, Meta-World, and VLABench extend coverage across multi-task manipulation, procedurally generated scenes, and language-conditioned reasoning [157]–[161].

Other benchmarks target particular evaluation gaps. SimplerEnv constructs control-matched simulations of two physical robot platforms [113]. THE COLOSSEUM applies systematic perturbations to evaluate generalization [162], while FurnitureBench and BEHAVIOR-1K broaden evaluation toward assembly and household activity [163], [164]. RoboArena supports distributed real-robot comparison through A/B evaluation [165]. Large datasets such as Open X-Embodiment, DROID, BridgeData V2, AgiBot World, RH20T, and RoboMIND also provide held-out evaluation splits, although they are training corpora rather than controlled benchmarks [3], [108], [166]–[169].

World-model evaluation uses a different set of criteria. WorldModelBench assesses instruction-conditioned generation against physical principles [111], Physics-IQ isolates

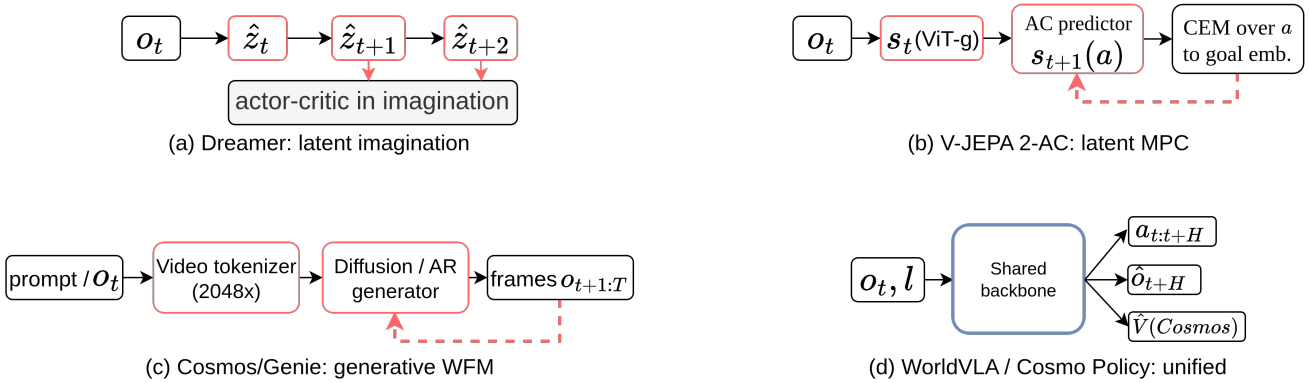


FIGURE 12. Architecture deep-dives across the world-model families: (a) latent-dynamics imagination (WM1), (b) latent-predictive planning (WM3), (c) generative world foundation models (WM2), and (d) unified world-action models (WM6).

TABLE 3. Limitation-to-candidate-mechanism matrix. The final column keeps *limitation evidence* separate from *mechanism-effect evidence*. Evidence labels are direct, indirect, adverse, or prospective; they describe the cited result, not an overall certainty grade or causal conclusion. Mechanisms remain candidates rather than established remedies (Sec. VI-I).

VLA limitation and its evidence	Candidate predictive mechanism (group)	Representative systems	Mechanism-effect evidence
L1 Interface does not expose candidate consequences; compounding under shift is established in imitation learning [107]	Action-conditioned prediction and candidate comparison (WM1, WM6)	WorldVLA; Cosmos Policy	*Indirect for WorldVLA:* auxiliary prediction improves configuration-specific LIBERO averages, but the attention-mask ablation addresses chunk decoding and candidate comparison is untested [16]. *Direct but narrow for Cosmos Policy:* a combined rollout-refinement and best-of-8 package improves mean completion by 12.5 points on two ALOHA tasks; it does not isolate search alone [17]
L2 Demonstration coverage and offline shift; interactive RL improves particular supervised baselines [11], [12]	Imagination RL and learned practice environments (WM1, WM5, WM6)	Dreamer family; World-Gymnast	*Direct but narrow/package-level.* World-Gymnast evaluates four real-robot tasks with 10 trials repeated five times: three tasks favor learned-world-model practice and close drawer favors the conventional simulator; generality and component attribution remain unresolved [147]
L3 Action-labeled robot data are costly; reported corpora use large teleoperation collections [3], [4]	Passive-video pretraining and synthetic data engines (WM2, WM3, WM5)	V-JEPA 2/2-AC; DreamGen	*Direct* for matched synthetic augmentation (13.1% to 38.3%) [14], [101]. *Indirect* for passive video: the base/adaptation data scales differ, but no reduced-video ablation isolates attribution [13]
L4 Physical/causal grounding is inadequately measured in policies; video benchmarks expose inconsistencies [110]; WorldModelBench remains protocol-unresolved [P] [111]	3D/contact-grounded state and intervention tests (WM4)	TesserAct; GWM	*Indirect/prospective.* The nonsignificant $r = -0.46$ is not evidence of no association, and the benchmarks test generators; no matched control study isolates improved mass, friction, or contact grounding [138]
L5 Static benchmarks saturate and robustness can collapse under perturbation [9]	Learned simulators and policy evaluators (WM2, WM5)	GE-Sim; Ctrl-World; WorldEval	*Adverse/indirect.* The human-scored Ctrl-World video-rollout pipeline in PolaRiS exhibits hallucinated interactions and policy mis-ranking [10]; reliable cross-policy rank transfer remains unestablished
L6 Objective/assurance gap: standard behavior cloning does not require calibrated risk or irreversibility estimates; abstention methods and dedicated safety benchmarks test missing properties [7], [114], [115]	Consequence, constraint, and uncertainty heads plus runtime shields (WM6; prospective WM4)	Cosmos Policy; safety benchmarks	*Prospective.* Task-value prediction is not calibrated safety or causal counterfactual evaluation; violation, irreversibility, and fallback metrics remain open [17]
L7 Prediction/search competes with deployment deadlines	Latent planning and shared predictive/action representations (WM1, WM3, WM6)	V-JEPA 2-AC; FLARE	*Indirect.* Roughly four minutes versus 16 seconds per planning query is an unmatched comparison with different candidates and representations [13]; FLARE ablates representation sharing, not online multi-candidate deliberation [148]

physical understanding using paired real videos [110], and VBench-2.0 evaluates intrinsic faithfulness in generated video [112]. EWMBench focuses more directly on embodied prediction [132]. WorldArena extends evaluation from perceptual quality to functional roles, including synthetic-data generation, policy evaluation, and action planning [170]. WorldArena 2.0 adds visuotactile observations, interactive reinforcement-learning environments, and simulated and

TABLE 4. Representative world models and predictive systems by overlapping taxonomy group. Year denotes the first public report. State: Pix = pixel/video, Lat = latent, 3D = geometry. A-cond: ✓ = executable-action conditioning, L = learned latent actions, J = joint future video/action generation, (✓) = game/interface controls; — = no such conditioning. Int: the prediction module is queried during closed-loop control, learned simulation, or evaluation; this does not by itself imply candidate-action planning. Open: ✓ = verified official public code or weights; — = no verified release. Task-conditioned prediction and auxiliary pretraining are distinguished from action-conditioned transitions. Results use each source's own protocol.

Model	Year	Grp	State	A-cond	Int	Open	Headline result (own protocol)
World Models [118]	2018	WM1	Lat	(✓)	✓	✓	Policy learning in imagined CarRacing/Doom environments
PlaNet [119]	2018	WM1	Lat	✓	✓	✓	Latent MPC from pixels
DreamerV3 [120]	2023	WM1	Lat	✓	✓	✓	150+ tasks, one configuration; Nature publication in 2025
TD-MPC2 [122]	2023	WM1	Lat	✓	✓	✓	80/104 tasks, single 317M agent
Dreamer 4 [15]	2025	WM1	Lat	(✓)	✓	—	Offline imagined-policy training; Minecraft diamond acquisition
UniPi [123]	2023	WM2	Pix	—	—	—	Text-conditioned video plus inverse dynamics
UniSim [124]	2023	WM2	Pix	✓	✓	—	Interactive learned simulator
GR-1 [129]	2023	WM2	Pix	—	—	✓	CALVIN ABCD→D: 94.9% first-task success; 4.21 chain length
Cosmos [109]	2025	WM2	Pix	—	—	✓	Open WFM platform, 2048× tokenizer
Genie 3 [131]	2025	WM2	Pix	(✓)	✓	—	Real-time 720p worlds, minutes [R]
Genie Envisioner [132]	2025	WM2/5	Pix	✓	✓	✓	GE-Act solves tasks where VLAs: 0%
V-JEPA 2 [13]	2025	WM3	Lat	—	—	✓	Action-free video representation pretraining
V-JEPA 2-AC [13]	2025	WM3	Lat	✓	✓	✓	Image-goal Franka planning; 16 s/query; 10 trials/skill/config.
VLA-JEPA [137]	2026	WM3/6	Lat	L	—	✓	Latent-action predictive pretraining for robot policies
LaWAM [136]	2026	WM3/6	Lat	L	✓	✓	Predicted latent subgoals; LIBERO 98.6%; 187 ms/chunk
TesserAct [138]	2025	WM4	3D	—	—	✓	Text-conditioned RGB-D-normal futures and inverse dynamics
GWM [139]	2025	WM4/5	3D	✓	✓	✓	Gaussian-state neural simulation for model-based policy optimization
DreamGen [14]	2025	WM5	Pix	—	—	✓	Synthetic video plus inverse-dynamics labels
GE-Sim [132]	2025	WM5	Pix	✓	✓	✓	Closed-loop neural simulator
WorldEval [146]	2025	WM5	Pix	✓	✓	✓	Hardware-free policy ranking [P]
UnifoLM-WMA-0 [151]	2025	WM5	Pix	✓	✓	✓	Open humanoid WM stack (no quant.)
World-Gymnast [147]	2026	WM5	Pix	✓	✓	✓	Four-task real-robot transfer: three positive and one adverse simulator comparison
DreamDojo [152]	2026	WM2/5	Pix	✓	✓	✓	44k h human-video pretraining; robot-action simulation, evaluation and planning
WorldVLA [16]	2025	WM6	Pix+act	✓	—	✓	Auxiliary action-conditioned prediction improves LIBERO policy ablations
FLARE [148]	2025	WM6	Lat	✓	—	✓	Up to +26%; action-free co-training
DreamVLA [149]	2025	WM6	Pix+lat	—	✓	✓	Predicted dynamics/depth/semantics condition action diffusion
Cosmos Policy [17]	2026	WM6	Pix+act	✓	✓	✓	Direct policy: LIBERO 98.5%; rollout-refinement + best-of-8: +12.5 points on two ALOHA tasks
DreamZero [153]	2026	WM6	Pix+act	J	✓	✓	14B joint video–action model; Flash: ∼7 Hz chunk inference (2 GB200s)
StructVLA [150]	2026	WM6	Pix+act	—	—	✓	Sparse visual planning pretraining followed by action-policy tuning
HarmoWAM [154]	2026	WM6	Pix+act	—	✓	✓	Video foresight with adaptive transit/manipulation action experts

physical platforms across multiple embodiments [171].

Only a few recent benchmarks begin to connect the two evaluation traditions. ManipArena evaluates VLA and world-model approaches under a shared real-world protocol with synchronized real-to-simulation environments [172]. WorldArena and WorldArena 2.0 assess world models according to their functional contribution to learning and control [170], [171]. These benchmarks narrow the divide, but they remain recent and have not yet established a common evaluation standard.

The underlying issue is that four distinct notions of world-model quality are often conflated. A model may produce realistic videos without preserving physical dynamics or ranking candidate actions correctly. Conversely, an abstract latent model may support effective decisions without generating visually realistic observations. Table 5 separates these notions.

Table 6 summarizes the resulting evaluation landscape. Its purpose is to identify complementary coverage and unresolved gaps, not to imply that results obtained under different protocols are directly comparable.

B. COMPARISON PRINCIPLES

The available results support descriptive synthesis rather than statistical meta-analysis. Evaluations differ in embodiment, control frequency, observation inputs, training data, task splits, and reporting practice. They also rarely provide common effect measures, confidence intervals, or episode-level uncertainty. Pooling their headline scores would therefore produce a precision that the underlying evidence does not support.

Results are compared only when their experimental conditions can be established. Scores obtained from different benchmarks are not used to rank systems. When configurations differ within the same benchmark, they are reported separately. This distinction is important for LIBERO, where camera inputs, proprioception, data filtering, training procedures, and suite coverage vary across studies. A shared

TABLE 5. Distinct notions of world-model quality. Each requires a different evaluation instrument, and strong performance on one dimension does not imply strong performance on another.

Notion	What it measures	Evidence of decoupling
Visual fidelity	Perceptual realism of generated rollouts	An eight-model comparison found a negative but nonsignificant association between MLLM-based visual judgments and physical understanding ($r = -0.46$, $p = .249$); FVD was not examined [110]
Physical fidelity	Consistency with geometry, contact, and mechanics	Physically inconsistent events can occur in visually plausible generations [111]
Decision utility	Whether predictions rank candidate actions or policies correctly	Latent models can support planning without generating pixels [13]; WorldArena identifies a perception–functionality gap across 14 world models [170]
Control performance	Task success when the model is coupled to a controller	Prediction quality does not necessarily transfer to manipulation success; gains have been demonstrated only for particular joint prediction–action systems [16], [17]

benchmark name consequently does not imply a matched protocol.

The evidence tags introduced in Table 2 remain visible throughout the comparison. They identify primary-source traceability, vendor reporting, and unresolved protocol details, but they are not interpreted as formal certainty grades. No confidence intervals or significance tests are inferred when the original study does not report the information required to calculate them.

C. SHARED-BENCHMARK RESULTS

Table 7 presents reported LIBERO results under their respective configurations. The scores illustrate progress within the benchmark family, but they should not be interpreted as a unified leaderboard.

The OpenVLA baseline reports a four-suite average of 76.5% [2]. OpenVLA-OFT reports substantially higher performance after changing the training data, decoding method, and observation configuration [8]. Its 95.3% and 97.1% averages correspond to different visual and proprioceptive inputs and must therefore remain separate. The π_0 result reported in the same study uses the modified data configuration and reaches 94.2% [8].

SimpleVLA-RL evaluates reinforcement-learning post-training only on LIBERO-Long. Its 91.7% one-demonstration result and 98.5% full-trajectory result are informative within those settings but cannot be compared directly with four-suite averages [11]. Cosmos Policy reports a 98.5% four-suite average in direct-policy mode under its own filtered-data and observation configuration [17]. Establishing superiority among these systems would require matched training data, observations, task episodes, and statistical reporting.

Evidence from other benchmarks is less complete. On SimplerEnv, discrete-diffusion decoding reports 71.2% visual-matching performance for the Google Robot setting and 49.3% overall performance for the Bridge setting [44]. Memory-augmented policies report results at a similar level, although unresolved protocol details prevent a definitive comparison [173]. On the CALVIN ABC→D split, GR-1 reports an increase from 88.9% to 94.9% after video-generative pretraining [129]. These findings suggest benefits within their respective settings, but they do not support cross-benchmark ranking.

D. CROSS-FAMILY PROFILES

Table 8 summarizes representative evidence from the VLA and world-model families. The profiles illustrate different forms of contribution, including benchmark performance, real-world deployment, openness, efficiency, and functional prediction. Because these criteria are not measured on a common scale, the table is an evidence map rather than a ranking.

The profiles reveal an asymmetry in the available evidence. VLA studies provide numerous task-success results but rarely isolate the effect of consequence prediction. World-model studies provide increasingly detailed measures of prediction and simulation quality, yet comparatively few demonstrate improvements in physical manipulation. The strongest cross-family evidence concerns auxiliary predictive training and synthetic trajectory generation. Evidence for candidate-action comparison, policy evaluation inside learned environments, and real-time predictive control remains narrow.

E. INTERPRETATION

The evidence cannot be reduced to a single system score. Task success, robustness, physical deployment, computational cost, data requirements, openness, and independent replication answer different questions. They are also measured on incompatible scales and have application-dependent importance. Assigning review-derived weights would therefore create an artificial ranking unsupported by the available studies.

For practical selection, systems should first be screened against non-negotiable constraints such as safety, hardware availability, control latency, and access. The remaining candidates can then be compared dimension by dimension under matched evaluation conditions. For research evaluation, the more important requirement is a protocol that measures prediction quality and manipulation performance within the same experiment.

The principal finding is therefore methodological as well as technical. Current benchmarks can establish whether a policy completes a task or whether a model generates a plausible future, but they seldom establish whether the predicted future supports a better decision. Progress at the intersection of VLAs and world models requires joint evaluation of physical validity, uncertainty calibration, candidate ranking, control latency, and real-robot success.

TABLE 6. Benchmark landscape for VLA policies and world models. Type: S = simulation, R = real, and S/R = matched simulation and physical evaluation. Horizon: S = single-stage and L = multi-stage.

Benchmark	Year	Type	Hor.	Primary focus	Principal caveat
VLA and policy benchmarks					
LIBERO [155]	2023	S	S/L	130 language-conditioned tasks in four suites	High reported scores; sensitivity to perturbations [9]
LIBERO-Plus [9]	2025	S	S/L	Robustness across seven perturbation axes	Restricted to the scope of the underlying LIBERO tasks
SimplerEnv [113]	2024	S/R	S	Control-matched simulation of two physical platforms	Limited embodiment coverage
CALVIN [156]	2022	S	L	Long-horizon sequences and ABC→D generalization	Single simulated robot platform
RLBench [157]	2020	S	S/L	One hundred manipulation tasks	Dependence on scripted demonstrations
ManiSkill3 [158]	2024	S	S/L	GPU-parallel large-scale evaluation	Simulation-only physics
RoboCasa [159]	2024	S	L	Generated kitchen environments	Simulation and layout bias
Meta-World [160]	2019	S	S	Multi-task and meta-reinforcement learning	Limited visual diversity and saturation on some tasks
RoboArena [165]	2025	R	S/L	Distributed real-robot A/B evaluation	Throughput and variation across sites
OXE, DROID, BridgeData V2, and AgiBot World [3], [108], [166], [167]	2023–25	R	—	Training datasets with held-out evaluation splits	Not controlled benchmarks
World-model benchmarks					
WorldModelBench [P] [111]	2025	S/R	S	Instruction-conditioned generation and physical consistency	Evaluates generation rather than control
Physics-IQ [110]	2025	R	S	Physical understanding from paired videos	No action conditioning
VBench-2.0 [112]	2025	S/R	S	Intrinsic faithfulness, including physical behavior	Targets general video rather than robotics
EWMBench [132]	2025	S/R	S/L	Scene, motion, and semantic quality in embodied prediction	Closely associated with one system family
WorldArena [170]	2026	S/R	S/L	Perceptual and functional evaluation	Recent benchmark with a limited task source
WorldArena 2.0 [171]	2026	S/R	S/L	Visuotactile prediction, interactive learning, and multi-embodiment evaluation	Recent preprint without broad independent replication
ManipArena [172]	2026	S/R	S/L	Shared evaluation of VLAs and world models using real-to-simulation twins	Recent benchmark restricted to 20 tasks

TABLE 7. Reported LIBERO success rates (%) under protocol-specific configurations. Rows do not constitute a common leaderboard. Third denotes the third-person camera; wrist and proprio denote wrist imagery and proprioception. Dashes indicate unreported or inapplicable values.

Model	Reported configuration	Spatial	Object	Goal	Long	Avg
OpenVLA [2]	Published baseline	84.7	88.4	79.2	53.7	76.5
OpenVLA-OFT [8]	Filtered training set; third-person image and language	96.2	98.3	96.2	90.7	95.3
OpenVLA-OFT [8]	Filtered training set; third-person and wrist images, optional proprioception, and language	97.6	98.4	97.9	94.5	97.1
π_0 [8]	Result reported by the OFT study using its modified training and observation configuration	96.8	98.8	95.8	85.2	94.2
SimpleVLA-RL [11]	One demonstration per task; post-reinforcement-learning result	—	—	—	91.7	—
SimpleVLA-RL [11]	Full-trajectory supervised and reinforcement-learning configuration	—	—	—	98.5	—
Cosmos Policy [17]	Direct parallel decoding; filtered policy data and unfiltered world/value training data	98.1	100.0	98.2	97.6	98.5

VIII. RESEARCH DIRECTIONS

The evidence synthesized in Table 3 and Section VII identifies ten research directions for integrating predictive models with robotic control. These directions span model architecture, training data, evaluation, deployment, and safety. They are connected rather than independent: reliable deployment depends on predictive models that remain stable under interaction, evaluation protocols that measure decision utility, and assurance mechanisms that detect unsupported or unsafe predictions. Fig. 13 organizes these dependencies by readiness stage; it does not imply specific completion dates.

A. PREDICTIVE FOUNDATIONS

Four directions concern the predictive substrate itself: unified prediction and action, control-oriented objectives, long-horizon memory, and physically grounded state representations.

D1: Unified prediction and action. World-action models increasingly use a shared architecture to predict environmental change and generate robot actions. Existing systems represent this integration through shared token vocabularies [16], alignment with future latent states [148], or adaptation of a pretrained world foundation model into a policy [17]. However, the conditions under which predictive supervision improves control remain unclear. Joint training may produce transferable representations, but it may also introduce competition for model capacity or optimization interference between prediction and action objectives.

Future work should isolate these effects through matched ablations of direct policies, auxiliary prediction, and ex-

TABLE 8. Representative evidence profiles across VLA and world-model families. Selection criteria differ across rows; the table therefore supports orientation rather than ranking. [R] and [P] identify vendor-reported and protocol-unresolved evidence.

Group	Representative evidence	Related system	Basis and scope
A Action tokens	OpenVLA-OFT [T]	SpatialVLA	LIBERO averages of 95.3% and 97.1% under different observation configurations; open weights and code [8]
B Continuous actions	$\pi_{0.5}$ [T]	π_0 / GR00T N1	Open weights and deployment in previously unseen homes [52]; GR-2 reports 97.7% under a gated, vendor-reported setting [105]
C Hierarchical reasoning	Hi Robot / ECoT [T]	OneTwoVLA	Open evidence for hierarchical execution and intermediate reasoning; Gemini Robotics 1.5 remains vendor-reported [7]
D Spatial grounding	SpatialVLA [T]	GraspVLA	Source-traced gains from spatial action tokens; GraspVLA addresses the narrower grasping problem [72]
E Interactive learning	SimpleVLA-RL [T]	RIPT-VLA	LIBERO-Long improvement from 17.3% to 91.7% in the one-demonstration setting; a separate full-trajectory configuration reaches 98.5% [11]
F Efficient deployment	SmolVLA [T]	BitVLA	Competitive performance with consumer-GPU training and reduced computational requirements [87]
G Embodiment	$\pi_{0.5}$ [T]	GR00T N1.5	Open evidence for long-duration mobile manipulation [52]; Helix remains vendor-reported
WM1 Latent dynamics	Dreamer 4 [T]	TD-MPC2	Offline imagined learning in Minecraft [15]; broad latent-control evidence from TD-MPC2 [122]
WM2 Generative video	Cosmos [T]	Genie Envisioner	Open platform with source-traced licensing and technical documentation [109]; Genie 3 interactivity remains vendor-reported [131]
WM3 Latent prediction	V-JEPA 2-AC [T]	DINO-WM	Image-goal Franka planning without task-specific demonstrations; evaluation uses ten trials per skill and configuration [13]
WM4 Physical state	No comparable profile	TesserAct / GWM	Relevant systems exist, but no matched control comparison was identified
WM5 Learned environments	DreamGen [T]	GE-Sim	Quantified gains from generated training trajectories [14]; evidence for broader learned simulation remains limited
WM6 World-action models	Cosmos Policy [T]	WorldVLA [T]	Direct-policy LIBERO average of 98.5% and RoboCasa result of 67.1%; a separate computationally intensive planning package improves two ALOHA tasks by 12.5 points [17]. WorldVLA reports gains from auxiliary predictive training [16]

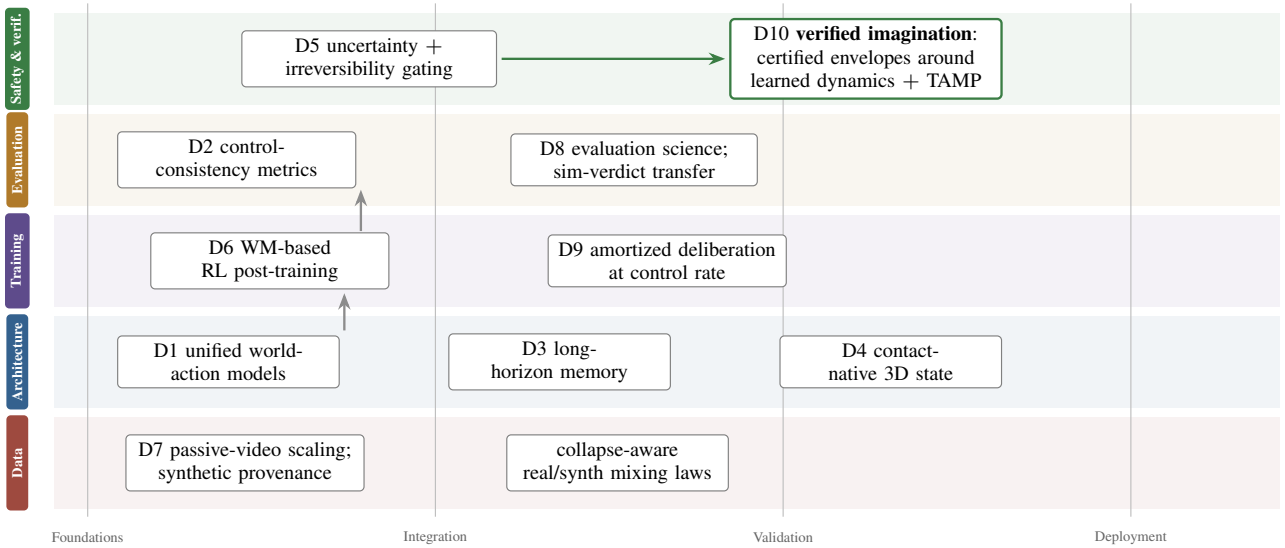


FIGURE 13. Illustrative dependency and readiness roadmap. The ten directions of Sec. VIII (D1–D10) are arranged in five layers; arrows mark dependencies, not dates. D10 asks when learned rollouts can be used inside a formally specified runtime assurance and task-and-motion planning stack.

plicit rollout-based planning. Important questions include how capacity should be distributed among prediction, action, and value heads; whether joint objectives obey identifiable scaling relationships; and when general world-foundation backbones such as Cosmos [109] complement or replace purpose-built VLA architectures.

D2: Control-oriented objectives. Perceptual realism is an incomplete objective for predictive control. Physics-IQ reports a negative but statistically nonsignificant association between its MLLM-based realism judgment and physical-understanding score across eight models ($r = -0.46$, $p = .249$) [110]. This result neither establishes independence between the two measures nor evaluates FVD, but it illustrates why appearance alone cannot establish decision utility.

A control-oriented model must respond correctly to changes in the applied action and preserve distinctions that affect task outcomes. Training objectives should therefore measure action sensitivity, state-transition consistency, candidate ranking, and closed-loop utility. EWMBench, WorldArena, and WorldArena 2.0 provide foundations for this direction by extending evaluation toward functional prediction, visuotactile interaction, and reinforcement-learning en-

vironments [132], [170], [171]. The remaining challenge is to develop differentiable control-consistency objectives that can be applied during training and shown to improve physical manipulation.

D3: Stable memory. Long-horizon prediction remains vulnerable to accumulated model error and loss of persistent state. Interactive consistency has been demonstrated for only limited durations, and some of the strongest claims remain vendor-reported [131]. Manipulation adds further difficulty because relevant objects may be occluded and contact state may not be recoverable from a single image.

Predictive systems therefore require memory that preserves object identity, geometry, task progress, and contact history across extended execution. Promising directions include retrieval-augmented prediction, explicit object-centric memory, and recurrent latent states with calibrated drift estimates. The gains reported for memory-augmented policies suggest that persistent context can already improve action generation [173]; equivalent evidence is still needed for world models. Evaluation should report prediction error as a function of rollout horizon and identify the horizon beyond which the model no longer supports reliable decisions.

D4: Contact-aware state. Physical interaction cannot be represented adequately through visual appearance alone. Force- and tactile-conditioned policies show that contact modalities provide useful control information [174], [175], while tactile world-action models extend predictive learning into contact-rich settings [142]. Nevertheless, this remains one of the least developed areas in the evidence map.

Progress requires representations in which force, compliance, friction, deformation, and contact mode are state variables rather than visual effects to be rendered. Hybrid neural–analytic dynamics and differentiable contact models may combine the adaptability of learned prediction with stronger physical structure. Corresponding benchmarks should include tasks whose success depends on predicting forces or material responses, so that appearance-based shortcuts cannot produce high scores.

B. LEARNING AND DATA

The next two directions concern the use of learned environments for policy improvement and the composition of the data used to train them.

D5: Imagined post-training. Reinforcement-learning improvements in VLA policies and advances in learned simulation suggest a common training loop: use a world model to generate interaction experience, optimize the policy within that environment, and validate the resulting behavior on the physical robot. World-Gymnast provides direct evidence that such transfer is feasible, although its comparison covers only four tasks and includes one task on which the conventional simulator performs better [147]. Dreamer 4 demonstrates large-scale offline learning in imagination in a different domain [15].

These results motivate world-model-based post-training but do not yet establish general robotic effectiveness. The main challenges are constructing useful reward signals, detecting policies that exploit model errors, and determining when imagined improvement transfers to physical execution. Learned practice may reduce hardware interaction, but some real-world validation remains necessary to identify model bias and update the operating region.

D6: Data composition. Passive video and synthetic trajectories could reduce dependence on costly robot demonstrations. Their contribution, however, depends on how they are combined with action-conditioned physical data. Passive video does not directly reveal interventions, while synthetic trajectories may transmit the biases of the model that generated them. Recursive use of generated data can also degrade the learned distribution, as demonstrated more generally in research on model collapse [176].

Future studies should report the provenance of real, passive, and synthetic samples separately. Controlled experiments are needed to estimate useful real-to-synthetic mixing ratios, determine when additional generated data cease to improve performance, and measure contamination across successive training generations. Scaling claims based on egocentric human video also require independent evaluation in dexterous robotic tasks; current evidence remains partly vendor-reported [101].

C. EVALUATION AND DEPLOYMENT

Predictive models will have limited practical value unless their functional contribution can be measured and their inference cost fits the robot's control requirements.

D7: Evaluation validity. Evaluation should establish whether a predicted future supports a better decision, not merely whether it looks plausible. This requires joint measurement of prediction accuracy, action ranking, policy success, and transfer to physical execution. Real-to-simulation correlation audits such as PolaRiS [10], distributed physical comparison through RoboArena [165], and versioned evaluation harnesses [177] provide useful components of such a framework.

A stronger evaluation standard should document the task version, embodiment, observation inputs, controller frequency, random seeds, episode counts, and uncertainty estimates. It should also report whether a learned simulator preserves policy rankings observed on the real robot. Power analysis and confidence intervals are particularly important because physical experiments often use small numbers of trials. The central unresolved question is when a simulator or world model is sufficiently reliable to support a decision about policy quality.

D8: Efficient deliberation. Explicit prediction introduces a trade-off between decision quality and response time. The reported V-JEPA 2-AC planning query requires approximately 16 seconds [13], whereas many manipulation tasks require much faster feedback. The acceptable latency depends on whether prediction occurs at the servo, skill, or task level, but current planning costs generally preclude frequent low-level replanning.

Several strategies could narrow this gap. Latent model-predictive control can reduce the cost of generating detailed observations, while learned proposal distributions can focus search on promising candidates [122]. Plans generated through slow deliberation can also be distilled into a fast policy, allowing the world model to provide supervision without running a full search at every step. An anytime architecture could allocate little computation to familiar states and increase deliberation when uncertainty or predicted risk rises. Evaluation should report planning latency separately from action playback and low-level controller frequency.

D. SAFETY AND ASSURANCE

Safety requires more than accurate average prediction. A system must recognize uncertainty, estimate the severity and reversibility of potential failures, and enforce constraints when its predictions are unreliable.

D9: Risk-aware control. Current evidence does not establish a complete manipulation system that combines consequence prediction with calibrated uncertainty, abstention, runtime protection, and a tested fallback controller. These functions should be evaluated separately. Epistemic uncertainty indicates insufficient model knowledge, aleatoric uncertainty reflects variability in possible outcomes, and constraint risk depends on both the probability and severity of failure. Irreversibility adds another dimension because some prediction errors can be corrected while others cause permanent damage.

Relevant components already exist. Conformal methods can trigger help-seeking or abstention [115], while control barrier functions and reachability analysis can shield the system from unsafe actions [116], [117]. World-action models can provide candidate consequences for inspection. Their integration remains an open problem: a world model alone does not constitute a safety case, and uncertainty estimates must be validated against actual physical outcomes.

E. VERIFIED PLANNING

D10: Verified imagination. The most ambitious direction is to combine learned world models with the explicit reasoning and assurance mechanisms of task-and-motion planning (TAMP). Classical TAMP provides effect models, feasibility tests, backtracking, and, under stated assumptions, formal properties such as soundness or completeness. Learned world models offer complementary capabilities: they can acquire predictive structure from data in open-world, contact-rich settings where analytic models are incomplete.

The reviewed manipulation literature does not yet demonstrate a system that uses a learned world model as the effect model inside a symbolic–geometric planner and verifies the resulting composition. This is a conclusion about the reviewed literature rather than a claim that no related work exists elsewhere. A credible formulation must state the conditions under which learned predictions are trusted.

Such a system requires four elements: an observation-to-state mapping with characterized estimation error; a mapping from predicted states to the physical predicates used by the planner; bounded transition error within a declared operating region; and support tests that prevent the planner from relying on predictions for unsupported states or actions. Reachable sets, chance constraints, or barrier certificates could then bound the predicted trajectories [116], [117]. An out-of-distribution detector and runtime shield would route unsupported or unsafe actions to a conservative fallback.

Language- or vision-language-based task planners [66] could query this verified predictive interface when testing action effects rather than treating generated rollouts as factual. Any claims of soundness or completeness would remain conditional on the predicate abstraction, model-error bounds, search procedure, and fallback policy. Evaluation should report task success together with constraint violations, false-safe and false-unsafe decisions, coverage of the verified operating region, intervention frequency, and missed computational deadlines. The resulting guarantees would belong to the complete planning and assurance architecture, not to the learned world model in isolation.

Outlook. The ten directions form a progression from predictive capability to dependable deployment. Unified architectures, control-oriented objectives, stable memory, and contact-aware representations provide the predictive foundation. Imagined post-training and mixed data sources address learning scale, while functional evaluation and efficient deliberation determine whether prediction is useful in practice. Risk-aware control and verified planning then place learned prediction within an explicit assurance framework.

The highest-priority unresolved problem is the composition of these capabilities: consequence-aware prediction, calibrated uncertainty, runtime gating, and safe fallback behavior within one evaluated manipulation system. As a forward-looking position, we expect the boundary between VLAs and world models to become less architectural and more functional. Prediction and action may increasingly share a backbone, while remaining separable as objectives and evaluation targets.

IX. CONCLUSION

This review characterizes the movement from direct vision–language–action policies toward predictive systems for robotic manipulation. Its main contributions are a faceted VLA description, an overlapping taxonomy of predictive systems, and a matrix that separates evidence for seven limitation and evidence-gap categories from evidence that a proposed mechanism changes them. This separation matters: the limitations arise at different levels—interface, objective, data, representation, evaluation, assurance, and deployment—and none establishes that a world model is necessary or sufficient.

The most direct matched mechanism-effect evidence in the reviewed manipulation corpus concerns synthetic-trajectory augmentation. Auxiliary predictive supervision also improves particular world-action policies. Direct candidate-

comparison evidence is narrow: the Cosmos result combines rollout refinement with best-of-8 selection on two ALOHA tasks rather than isolating search alone. V-JEPA 2's action-free pretraining and V-JEPA 2-AC's action-conditioned adaptation illustrate passive-video leverage but do not provide a controlled sample-efficiency substitution. Cosmos Policy reports a 98.5% LIBERO average in direct-policy mode; its 12.5-point planning-package gain is a separate two-task experiment. These results show that predictive representations can coexist with strong policy performance, not that a common cross-system ranking or a general planning advantage has been established. World-Gymnast adds direct but narrow evidence for real-robot transfer after RL in a learned practice environment: three of four task comparisons favor the learned-world-model package and one favors the conventional simulator, so neither universal benefit nor component-level attribution follows.

The priority is therefore measurement and integration: action-conditioned intervention tests, matched data ablations, physics and contact perturbations, learned-evaluator validity, calibrated uncertainty, and explicit separation of servo, policy-query, and replanning rates. For safety-critical use, predictions must sit inside a specified assurance stack with distribution-shift detection, constraints, monitoring, and fallback control. Verified imagination is a research program rather than a current capability: its guarantees must be stated relative to observable-to-latent mappings, model-error bounds, physical predicates, and runtime enforcement. This evidence-calibrated agenda is more defensible than forecasting a fixed timetable or treating prediction itself as proof of causal understanding, safety, or deployment readiness.

APPENDIX. TRAINING CORPORA AND LICENSES

Table 9 summarizes the corpora referenced throughout Sections IV–VII. Sizes and licenses are as reported by each corpus's primary publication or official release page at extraction time (August 2026); licenses should always be confirmed against the official distribution, which governs, before any reuse in derivative work.

TABLE 9. Major training corpora for VLA and world-model research. Sizes and licenses as reported by each corpus's primary publication or official release page at extraction time (August 2026); the official distribution governs in all cases and should be confirmed before reuse.

Corpus	Reported scale	Modality	License
Open X-Embodiment [3]	970k+ trajectories, 22 embodiments	Robot traj.	Per-dataset (mixed)
DROID [108]	76k episodes, 564 scenes	Robot traj.	CC-BY-4.0
BridgeData V2 [166]	60k+ trajectories	Robot traj.	CC-BY
AgiBot World [167]	1M+ trajectories, 100+ robots	Robot traj.	CC-BY-NC-SA-4.0
LeRobot [178]	Community aggregation (growing)	Robot traj.	Apache-2.0 (code); per-dataset licenses vary
Ego4D [179]	3,670 h egocentric video	Human video	Ego4D license
VideoMix22M [13]	22M videos / 1M+ h	Internet video	Per-source (mixed)
DreamGen traj. [14]	Synthetic neural trajectories	Generated	Apache-2.0

DATA AVAILABILITY

The primary-source-verified quantitative data underlying Figs. 6 and 10 will be released in the project repository at `[repository URL pending]` as `VLA_WM_verified_data_`. The workbook contains only the retained, verified fields used in these figures and preserves each value's measurement scope, unit, exact/approximate/bound semantics, source title and version, source URL and locator, verification status, and row- and column-level audit decision. Unresolved or unauthenticated source columns are excluded from the verified plot tables and documented in the accompanying audit sheets rather than presented as evidence. The same release will include machine-readable verified CSV files, vector and raster figure outputs, a SHA-256 manifest, dependency information, and the Python scripts `validate_verified_package.py` and `generate_verified_plots.py`. These source-reported quantities have heterogeneous measurement scopes and should not be interpreted as a common sample-efficiency or benchmark leaderboard.

CONFLICT OF INTEREST

AUTHOR CONTRIBUTIONS

Muhayy Ud Din: conceptualization; systematic database search and deduplication; title/abstract screening; full-text screening of the VLA corpus; taxonomy design; data extraction and provenance annotation across the six dimensions of Section VII-B ([T]/[R]/[P]); writing—original draft. **Ahsan B. Bakht:** title/abstract screening; full-text screening of the world-model and benchmark/evaluation corpora; data extraction; benchmark tables and figures; writing—original draft (Sections VI–VII). **Lakmal Seneviratne:** supervision; audit of the included corpus and exclusion log; writing—review and editing. **Irfan Hussain:** supervision; arbitration of screening disagreements; independent audit of [T]/[R] tag assignments; funding acquisition; writing—review and editing.

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Generative-AI disclosure: generative AI was used for language editing and, as documented in Section III-B, to suggest candidate records. Suggested records were screened under the same criteria and verified against primary sources; model output was not treated as evidence. The authors reviewed the manuscript and accept full responsibility for its content.

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